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REVIEW

A Closer Look into Recent Video-based Learning Research: A Comprehensive Review of Video Characteristics, Tools, Technologies, and Learning Effectiveness

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Abstract

People increasingly use videos on the Web for learning, be it for daily tasks in formal or informal educational settings. To enhance this type of learning, scientists are continuously conducting experiments, proposing guidelines, analyzing data, and researching artificial intelligence methods for tool development. In this comprehensive review, we analyze 257 articles (using PRISMA guidelines) on video-based learning from a technological perspective for the period from 2016 to 2021. One of the aims is to identify video characteristics that support learning as explored by previous research. Based on our analysis, we suggest a taxonomy that organizes the video characteristics and contextual aspects into eight categories: (1) audio features, (2) visual features, (3) textual features, (4) instructor behavior, (5) learners' activities (play, pause, etc.), (6) interactive features (quizzes, etc.), (7) production style, and (8) instructional design. Also, we identify four representative methodological approaches: (1) tool support of video-based learning, (2) controlled experiments, (3) data analysis studies, and (4) design guidelines for learning videos. We find that the most explored characteristics are textual features followed by visual features, learners' activities, and interactive features. Tools that aid learning through videos frequently utilize text from transcripts, video frames, and images. The learner's activity is heavily explored through log files in data analysis studies, and interactive features are frequently scrutinized in controlled experiments. As further contributions, we contrast research findings on how video characteristics affect learning effectiveness, report on tasks and technologies used to develop tools, and summarize design guideline trends to produce learning videos. Our findings provide actionable insights for the design of intelligent educational systems that better support video-based learning.

Keywords Video-based learning · Web-based learning · Video features

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Introduction

Video-based learning (VBL) is a term that has been adopted in several studies (e.g., Giannakos, 2013; Poquet et al., 2018; Yousef et al., 2014; Sablić et al., 2021). It refers to the task of learning using video sources, that is, acquiring knowledge and skills through video materials. In June 2022, reports stated that about 500 hours of new footage was uploaded every minute on YouTube (2025), with around 50% of views linked to learning purposes (Smith et al., 2018). This trend has only accelerated in recent years (YouTube, 2025), and global projections indicate that the number of YouTube users will increase by 232.5 million between 2024 and 2029, representing a growth of 24.91% (Global, 2025). With the ongoing proliferation of video resources, learners need to trust the expertise of the content creators and instructors in designing and producing effective learning videos. However, even on dedicated educational platforms, many videos are published without adherence to instructional design principles or established best practices (Guo et al., 2014). As research in multimedia learning has demonstrated, the effectiveness of VBL depends critically on the way video content is designed (Mayer, 2021). Although efforts to better understand VBL have increased (Poquet et al., 2018; Giannakos, 2013; Yousef et al., 2014), essential questions remain unanswered, for example: What are the characteristics of a video[1] that lead to optimal learning, according to individual needs, the learning goals, and contextual aspects? Complementing the focus on effective design of instructional videos, Artificial Intelligence (AI) technologies are increasingly being developed to enhance VBL effectiveness. These AI methods aim at automating key tasks such as indexing, processing, enrichment (e.g., by linking other sources), providing intelligent feedback, summarizing information, recommending additional videos, or adapting the learning environment to individual needs.

Our taxonomy of video characteristics-particularly the categories of interactive features, learner activities, and instructional design-provides a critical foundation for developing more sophisticated adaptive learning systems. For instance, understanding how visual and textual elements affect different learner types enables AIED researchers to tailor video content to individual profiles. Similarly, the patterns of learner activities we identify (play, pause, rewind) offer valuable behavioral cues that intelligent tutoring systems can leverage to assess comprehension and engagement in real-time, enabling more responsive feedback mechanisms. The data analysis studies in our review reveal how machine learning methods can process video interaction logs to identify struggling learners, predict performance, and generate personalized learning paths-key capabilities for next-generation AIED systems. Furthermore, our findings on interactive features (e.g., quizzes and annotations) offer guidance for designing feedback mechanisms at optimal intervention points. As VBL continues to evolve, it is crucial for AI-driven learning environments to apply these evidence-based insights for supporting personalized learning through adaptive content selection, intelligent feedback, and learner-responsive interventions-vital priorities for advancing AI in Education.

[1] In the context of this review, we understand video characteristics as specific attributes, features, or aspects of a video that might be relevant to VBL.

Previous VBL reviews (Poquet et al., 2018; Giannakos, 2013; Yousef et al., 2014) performed descriptive analysis on the growth of the field, its methodological approaches, the disciplines addressed by the videos, their production style, the mode of use (i.e., main or supplementary learning medium), the context (i.e., formal or informal learning), video characteristics, and other dimensions. However, most of them emphasized experimental research without systematically examining other key methodological approaches. Moreover, although these reviews acknowledged certain video characteristics, they did not delve further into technological perspectives, supportive tools, and didactic aspects in an integrated manner. Furthermore, they did not dive deeper into bridging these video characteristics under a comprehensive, technology-oriented taxonomy.

In this paper, we present a comprehensive review of 257 articles on video-based learning from computer science (CS) databases for the period from 2016 to 2021, collected using the PRISMA guidelines (Page et al., 2021). While databases in other fields might offer valuable pedagogical insights connected to VBL, we focus on CS databases to prioritize the technological view that aligns with our research goals and scope. Compared to other reviews, we consider the largest number of articles. Thus, we can achieve a robust analysis of the state of the art in the field. In contrast to previous reviews that focused on experimental research, we identify and consider four main types of methodological approaches: (1) proposals of tools to support VBL, (2) studies with controlled experiments, (3) data analysis studies, and (4) proposals of design guidelines for learning videos. Specifically, our research goals encompass a close investigation of: (a) methodological approaches, (b) video characteristics, (c) tasks and technologies used to develop methods, tools, and frameworks that support learning, (d) guidelines to design learning videos, as well as (e) findings from controlled experiments and data analysis. Furthermore, we derive a taxonomy that organizes video characteristics and contextual aspects into eight categories: (1) audio features, (2) visual features, (3) textual features, (4) instructor behavior, (5) learners' activities, (6) interactive features, (7) production style, and (8) instructional design.

The findings of this study can offer valuable insights to the education research community, particularly for researchers who are interested in AI-based support for learning with videos. The results are relevant for researchers who work on intelligent educational systems that include video lectures as well as MOOC platforms or video recommendation systems for educational purposes. The proposed taxonomy provides a holistic view and allows to systematically consider the critical components of a learning video. This approach can support the assessment, optimization, and customization of educational video content. Additionally, our review of tools, tasks, and technologies highlights actionable insights on how to incorporate artificial intelligence methods in video-assisted learning environments. Furthermore, our review on the effectiveness of characteristics and instructional design guidelines can support the (potentially AI-assisted) creation of video content that aligns with evidence-based findings. Finally, the identified trends and future directions can support the community in identifying further research gaps and challenges.

This review extends our previous work (Navarrete et al., 2021), where we reviewed VBL literature for the years 2020 and 2021. Here, we expand the reviewed period to

2016-2021 and also contrast and summarize the results of experimental research. We exclude research influenced by new advances in large language models and generative AI that emerged after our review period, thereby focusing solely on work predating these developments. By focusing on this time frame, we establish a comprehensive framework of video-based learning research that precedes the transformative impact of large language models and generative AI. This deliberate temporal boundary allows us to document the state of the field immediately before these technological paradigm shifts, providing a valuable baseline for future comparative research on how these advances have subsequently reshaped video-based learning approaches. Moreover, the developed taxonomy and findings serve as groundwork for next-generation learning systems that integrate generative AI capabilities while maintaining evidence-based principles of effective video-based learning.

The rest of the paper continues with a discussion of existing reviews on video-based learning (Section "Related Work"), a description of our methodology (Section "Methodology"), and a detailed analysis of the results of the review (Section "Literature Review"). After that, we discuss the impact of said video characteristics on learning effectiveness (Section "Impact of Video Characteristics on the Learning Effectiveness"). Finally, we draw conclusions and point out limitations and future work (Section "Conclusions").

Related Work

In this section, we establish the necessary foundations for our systematic review. We begin by contextualising video-based learning research, highlighting its relevance as an educational medium, examining frameworks for effective instructional video design, and situating VBL within the evolving landscape of artificial intelligence in education. Finally, we review existing systematic analyses of video-based learning literature to identify the gaps that our work addresses and to situate our contribution within the broader research landscape.

Contextualising Video-Based Learning Research

This review will focus on the technological aspects surrounding video-based learning, i.e., studies, tools and guidelines that focus on the support and improvement of VBL with technological means. This is, however, by far not the only research perspective one can apply with respect to VBL. While a detailed discussion of perspectives and methodological approaches is beyond the scope of this review, we will provide a short summary of relevant contextual dimensions.

Relevance of Video-based Learning

Empirical research has explored the advantages of video-based learning over other types of educational media. Aspects researched are, for instance, the flexibility provided by video as to control of pacing (Scagnoli et al., 2019), revisiting challenging parts of the content (Brecht & Ogilby, 2008), and the access to educational content regardless of location or time constraints. Studies have shown that well-designed

educational videos can improve knowledge retention (Mayer et al., 2008; Mayer & Anderson, 1991, 1992), increase student engagement (Mayer et al., 2004; Greenleaf et al., 2011) and provide more effective demonstrations of complex processes than static text or images alone (Mayer & Anderson, 1991, 1992).

Video-based Learning Design

The effectiveness of video-based learning depends critically on how videos are designed and implemented. The Cognitive Theory of Multimedia Learning (Mayer, 2014) provides an evidence-based framework that explains how learners process information from videos and identifies principles for optimizing cognitive load. Research applying this framework has established that videos incorporating principles such as segmentation, coherence, and signaling, significantly outperform those lacking these design elements (Mautone & Mayer, 2001; Moreno & Mayer, 2005; Mayer et al., 2017, 2019). Beyond cognitive considerations, frameworks addressing motivational aspects and social presence (Mayer, 2021) have further refined our understanding of what makes educational videos effective. These theoretical foundations inform the technological features we examine in our review.

Artificial Intelligence in Educational Contexts

Our review covers the critical period of video-based learning research immediately preceding the transformative emergence of large language models and generative AI technologies. The AI-based tools analyzed in our review represent the state of the field through 2021, capturing approaches to video analysis, recommendation, navigation, and interaction that form the technological foundation upon which current AI advancements are building. Recent developments are rapidly transforming the VBL landscape: large language models now enable automatic generation of educational video scripts (GitHub, 2025), real-time video summarization (Mu et al., 2024), and personalized question generation (Zanwar et al., 2024; Bhanot et al., 2023; Forkan et al., 2023); multimodal AI can analyze both visual and audio content simultaneously to identify learning gaps (Wang et al., 2024); and AI-powered virtual instructors can dynamically respond to learner questions about video content (Bodonhelyi et al., 2024). By documenting and organizing the video characteristics, methodological approaches, and technological approaches from the pre-generative AI era, our review provides a structured framework for understanding how these emerging AI capabilities might further enhance video-based learning. The taxonomy we develop and the patterns we identify serve as an essential groundwork for researchers exploring how newer AI technologies can address established challenges and extend the effectiveness of video-based learning systems.

Previous Video-based Learning Reviews

To the best of our knowledge, four survey articles on VBL have been presented previously. Table 1 shows the core information on these reviews. Giannakos (2013) focused on the number of publications on VBL, the shift of VBL research from theoretical

Table 1 Related survey articles: reference and year of publication, temporal scope of the survey, number of reviewed papers, and reviewed characteristics

Reference	Review period	N° papers	Reviewed dimensions	
Giannakos (2013)	2000-2012	166	1) Type of research	4) Technology type
			2) Sample characteristics	5) Style of use
			3) Subject area	
Yousef et al. (2014)	2003-2014	76	1) Learning effectiveness	3) Design
			2) Teaching methods	4) Reflection
Poquet et al. (2018)	2007-2017	178	1) Video type	4) Sample size
			2) Population	5) Video characteristics
			3) Control of prior knowledge	
Sablić et al. (2021)	2008-2019	39	Teacher perception and reflection	

to empirical, and the research type (e.g., theoretical, empirical), sample, discipline area, use of stable or mobile devices, and the context of the video usage (formal, and informal). Yousef et al. (2014) summarized the findings of experimental research on the effectiveness of VBL, teaching methods when using videos, integration of tools with VBL content, and the use of videos to reflect on teaching methods. Poquet et al. (2018) also focused on experimental studies. The authors presented a descriptive analysis of the types of videos, the population to which the experiment was directed, if the experiment controlled for prior knowledge, the sample size, and growth patterns in the number of publications. The authors also reviewed the video characteristics that influence learning effectiveness, especially the learning outcome. The most recent work was presented by Sablić et al. (2021), who reviewed research on the use of videos to support teachers' self-reflection and professional development. There are other reviews on learning from YouTube videos such as the recent work by Shoufan and Mohamed (2022). We have not considered this study or similar studies since they did not meet our criteria of focusing on VBL in general, and not only on a specific platform.

The reviews confirm the significant growth of research articles in the VBL field (Poquet et al., 2018; Giannakos, 2013; Yousef et al., 2014). Giannakos (2013) and Poquet et al. (2018) reported a noticeable increase of empirical and quantitative research over theoretical and qualitative. Furthermore, Giannakos (2013) showed a shift in the research domain, from the social sciences to technological domains, which was also confirmed by Poquet et al. (2018). Videos on science, technology, engineering, and mathematics (STEM) disciplines are currently a predominant research focus. Among non-STEM fields, psychology, humanities, and social sciences received the most attention. Other findings (Poquet et al., 2018; Yousef et al., 2014) suggested that the effectiveness of VBL has frequently been evaluated through two learning outcome metrics: recall (remembering information) and transfer (applying what was learned to

a different scenario). Other popular metrics of learning effectiveness were motivation, cognitive load, mental effort, attention, and affect (Poquet et al., 2018). The most commonly analysed data sources were clickstreams (e.g., views, likes) and eye-tracking data (Poquet et al., 2018). Regarding video characteristics, Poquet et al. (2018) found that the most examined were text, audio, animations, and voice over slides. Other investigated features included text customization (e.g., personal pronouns), signaling (e.g., highlight information), video usage (e.g., replacing face-to-face lectures) content (e.g., metaphors vs. descriptive language), task (e.g., annotation features) quizzes, learner characteristics, learner Control (e.g., pause, play), and distraction. Yousef et al. (2014) found that trending topics were the effects of annotations and authoring tools, collaborative VBL, and video usage for reflective teacher education.

Methodology

The methodology for our systematic review was inspired by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Page et al., 2021). This provides a framework to summarize results of research articles. The articles were retrieved from SpringerLink (Springer International Publisher Science, 2025), the Digital Library of the Association for Computing Machinery (ACM DL) (Association for Computing Machinery, 2025), and the Digital Bibliography and Library Project (DBLP) (Dblp, 2025). These are focused on computer science domains, and therefore, fit well our task of reviewing the technological aspects of VBL. We used advanced search options to refine our results in the databases. In ACM DL, we filtered the articles by matching the query terms in the title in the defined time frame. In SpringerLink, we applied the same filters plus an additional filter to obtain articles in CS. In DBLP, we did not apply filters since this covers already CS and matches the query term to the titles. Table 2 shows the set of query combinations in the order we used them, which includes the keywords from our previous work (Navarrete et al., 2021) plus new terms (massive open online courses *MOOC* and *Lecture*).

We included a paper in the final set when the article: (a) considered videos in academic scenarios (we excluded video tutorials for everyday tasks such as cooking), (b) focused on videos as learning resources (we excluded the use of class recordings for reflective tasks such as in teacher education), and (c) examined or leveraged specific

Table 2 This table shows the set of query terms and the order used to find relevant articles

Base	Extension				
Video	1.1) Education	1.2) Educational	2.1) Instructional	2.2) Instruction	3.1) Teach
	3.2) Teaching	4) -based Learning	5) Tutorial	6.1) Learning	6.2) Learn
	7.1) Explaining	7.2) Explanatory	8) Classroom	9) Student	10) MOOC
	11.1) Knowing	11.2) Knowledge	12) University	13) Skills	14) School
	15) Lecture	16) Pupil			

The base term in the first column was combined with each of the extensions in the second column (using logical AND as a connector)

Table 3 Taxonomy of video characteristics

Category	Sub-category	Examples
Audio features	Audio records (Shi et al., 2019; Mahapatra et al., 2018a; Ohnishi et al., 2019; Husain & Meena, 2019; Soares & Barrére, 2018; Kanadje et al., 2016; Gandhi et al., 2016; Kumar et al., 2017; Zhao et al., 2017; Mahapatra et al., 2018b; Bian & Huang, 2019; Balasubramanian et al., 2016; Walk et al., 2020; Furini, 2018; Jr. et al., 2019; Lin et al., 2019; Sharma et al., 2016b; Chatbri et al., 2017; Furini, 2021; Sanchez-Cortina et al., 2016; Gopalakrishnan et al., 2017)	
	Sound-related (Shoufan, 2019b; Ding, 2018; Shi et al., 2019; Shoufan & Mohamed, 2017; Mukherjee et al., 2019; Soares & Barrére, 2018; Kanadje et al., 2016; Ghauri et al., 2020; Furini, 2018; Imran et al., 2019; Chen et al., 2020)	Energy, entropy, jitter, noise, Mel-frequency cepstral coefficients, perceived quality
	Representation (cues) (Wang et al., 2020b; Shi et al., 2019)	Instructor-related: verbal cues
Visual features	Video frames and images (figures and illustrations) (Zhao et al., 2018b; Mahapatra et al., 2018a; Koka et al., 2020; Husain & Meena, 2019; Ma et al., 2017; Kumar et al., 2017; Zhao et al., 2017; Yadav et al., 2016; Mahapatra et al., 2018b; Bian & Huang, 2019; Zhao et al., 2018b; Ohnishi et al., 2019; Yadav et al., 2016; Xu et al., 2019b; Cagliero et al., 2019; Balasubramanian et al., 2016; Rahman et al., 2020; Ghauri et al., 2020; Khattabi et al., 2019; Jung et al., 2019; Davila et al., 2021; Kota et al., 2020; Guan et al., 2019; Kota et al., 2019; Xu et al., 2019a; Davila & Zanibbi, 2018; Furini, 2018; Lee et al., 2017b; Garber et al., 2017; Liu et al., 2017; Ponzanelli et al., 2019; Yadid & Yahav, 2016; Zhong & Ma, 2018; Alahmadi et al., 2018; Khandwala & Guo, 2018; Che et al., 2016; Lin et al., 2019; Jung et al., 2019; Guan et al., 2019; Furini et al., 2018; Waykar & Bharathi, 2017; Ciurez et al., 2019; Dessì et al., 2017; Nong et al., 2019)	
	Representation (cues) (Wang et al., 2020b; Moon & Ryu, 2021; Pi et al., 2019; Wang et al., 2019b; Pi et al., 2017b; Sharma et al., 2016c; Mahapatra et al., 2018a; Jung et al., 2018; Sharma et al., 2016c)	Instructor-related: pointing gestures, directed gaze. Content-related: arrows, moving objects
	Quality-related (Shoufan, 2019b; Ding, 2018; Shoufan & Mohamed, 2017)	Video quality, resolution in pixels

Table 3 continued

Category	Sub-category	Examples
Textual features	Transcripts (Shi et al., 2019; van der Sluis et al., 2016; Atapattu & Falkner, 2017; Mukherjee et al., 2019; Kravvaris & Kermanidis, 2019; Das & Das, 2020; Ohnishi et al., 2019; Husain & Meena, 2019; Soares & Barrére, 2019; Galanopoulos & Mezaris, 2019; Soares & Barrére, 2018; Gandhi et al., 2016; Kumar et al., 2017; Zhao et al., 2017; Yadav et al., 2016; Mahapatra et al., 2018a; Bian & Huang, 2019; Ma & Ma, 2019; Yoo et al., 2021; Vimalaksha et al., 2019; Balasubramanian et al., 2016; Kim & Kim, 2016; Ghauri et al., 2020; Liu et al., 2018; Jo et al., 2019; Jung et al., 2019; Walk et al., 2020; Kravvaris & Kermanidis, 2019; Furini, 2018; Albahr et al., 2021; Jr. et al., 2019; Ponzanelli et al., 2019; Escobar-Avila et al., 2017; Nangi et al., 2019; Che et al., 2016; Tavakoli et al., 2020; Bleoanca et al., 2020; Schulten et al., 2020; Jordán et al., 2020; Zhao et al., 2018a; Aytekin et al., 2020; Jung et al., 2018; Tang et al., 2021; Walk et al., 2020; Borges & dos Reis, 2019; Kravvaris & Kermanidis, 2017; Basu et al., 2016b; Cooper et al., 2018; Coccoli & Vercelli, 2016; Ji et al., 2018; Fraser et al., 2019; Stoica et al., 2019; Basu et al., 2016a; Dessì et al., 2017; Chatbri et al., 2017; Wang et al., 2020a; Horovitz & Ohayon, 2020; Hernandez & Yang, 2021)	
	Metadata (Eradze et al., 2020; Saurabh & Gautam, 2019; Shoufan, 2019b; Lee et al., 2017a; Silva & Azevedo, 2017; Hildebrand & Ahn, 2018; Sjödén et al., 2018; Meseguer-Martinez et al., 2019; Galster et al., 2018; Das & Das, 2020; Liu et al., 2018; Kravvaris & Kermanidis, 2019; Jr. et al., 2019; Ponzanelli et al., 2019; Escobar-Avila et al., 2017; Che et al., 2016; Tavakoli et al., 2020; Jordán et al., 2020; Zhao et al., 2018a; Tang et al., 2021; Basu et al., 2016b; Cooper et al., 2018; Borges & Silveira, 2019; Fraser et al., 2019; Stepanek & Dorn, 2017; Othman et al., 2016; Sung et al., 2016; Stoica et al., 2019; Mubarak et al., 2021; Bulathwela et al., 2020; Basu et al., 2016a; Poché et al., 2017; Othman et al., 2017; Lei et al., 2019)	Titles, keywords, tags, video length, comments, course syllabus, video url
	On-screen text (der Zee et al., 2017; Ozdemir et al., 2016; Whitney & Dallas, 2019; Zhao et al., 2018b; Mahapatra et al., 2018a; Ohnishi et al., 2019; Husain & Meena, 2019; Das & Das, 2020; Gandhi et al., 2016; Kumar et al., 2017; Zhao et al., 2017; Mahapatra et al., 2018a; Koka et al., 2020; Bian & Huang, 2019; Xu et al., 2019b; Cagliero et al., 2019; Balasubramanian et al., 2016; Davila et al., 2021; Kota et al., 2020; Xu et al., 2019a; Davila & Zanibbi, 2018; Lee et al., 2017b; Garber et al., 2017; Ponzanelli et al., 2019; Yadid & Yahav, 2016; Zhong & Ma, 2018; Alahmadi et al., 2018; Liu et al., 2018; Khandwala & Guo, 2018; Che et al., 2016; Furini et al., 2018; Waykar & Bharathi, 2017; Ciurez et al., 2019; Dessì et al., 2017; Nong et al., 2019)	
	Representation (cues) (Mahapatra et al., 2018a)	Font size, boldeness

Table 3 continued

Category	Sub-category	Examples
Instructor behavior	Body-related (Beege et al., 2020, 2019; Wang et al., 2019a; Horovitz & Mayer, 2021; Li et al., 2016; Ding, 2018; Tian & Bourguet, 2016; Chen et al., 2020; Xu et al., 2020; Godavarthi et al., 2020; Hoogerheide et al., 2018; Shoufan & Mohamed, 2017)	Gestures: hand strokes and waving. Facial emotions, body language, pose, clothes
	Speech-related (Horovitz & Mayer, 2021; Shoufan, 2019b; Ding, 2018; van der Sluis et al., 2016; Atapattu & Falkner, 2017; Shoufan & Mohamed, 2017; Patel et al., 2018; Mahapatra et al., 2018a; Soares & Barrére, 2019; Balasubramanian et al., 2016; Jo et al., 2019; Jung et al., 2019; Che et al., 2016; Jung et al., 2019; Ji et al., 2018; Bulathwela et al., 2020)	Tone, speech rate, intonation, lexical diversity, prosodic characteristics, readability, silence rate
	Attitudes (Shoufan, 2019b)	Enthusiasm
Learner's activity	Eradze et al. (2020), Stöhr et al. (2019), Wynn et al. (2019), Costley et al. (2021), Angrave et al. (2020), Li (2019), Rodríguez et al. (2018), Costley and Lange (2017), Giannakos et al. (2016), Hu et al. (2020), Uchidiuno et al. (2018), Kovacs (2016), van der Sluis et al. (2016), Shoufan (2019a), Lei et al. (2017), Yan and Baxter (2018), Dodson et al. (2018a), Atapattu and Falkner (2017), Patel et al. (2018), He et al. (2018a), Walk et al. (2020), Choi et al. (2019), Long et al. (2018), Zhang et al. (2018b), Bothe and Meinel (2020), Wachtler et al. (2016a), Mbouzao et al. (2019), Min et al. (2019), He et al. (2018b), Zhang (2017), Brinton et al. (2016), Mubarak et al. (2021), Jeon and Park (2020), Lee and Muldner (2020), Aouifi et al. (2021), Furini et al. (2021), Mbouzao et al. (2020), Lemay and Doleck (2020), Korösi et al. (2018), Othman et al. (2017), Lallé and Conati (2020), Saurabh and Gautam (2019), Shoufan (2019b), Lei et al. (2019), Meseguer-Martinez et al. (2019), Lee et al. (2017a), Shoufan and Mohamed (2017), Tavakoli et al. (2020), Kravvaris and Kermanidis (2017), Zhang et al. (2018a), Tang et al. (2021), Merkt et al. (2018), Schroeder et al. (2020), Garrett (2021), Lang et al. (2020), Bulathwela et al. (2020)	Explicit: play, pause, speed rate, seek forward. Implicit: viewing percentage, average of videos watched per week. Popularity-related: user ratings, views, likes. Pace-related: automatic pauses, learner-controlled pauses, fast pace
Interactive features	Kleftodimos and Evangelidis (2018), van Alten et al. (2020), Leisner et al. (2020), Keller et al. (2019), der Zee et al. (2018), Sözeri and Kert (2021), Taskin et al. (2019), Lu et al. (2021), Kuhail and Aqel (2020), Chen et al. (2021), Winkler et al. (2020), Wachtler et al. (2016b), Cojean and Jamet (2017), Ku et al. (2019), Palaigeorgiou and Papadopoulou (2019), Moos and Bonde (2016), Shelton et al. (2016), da Graça Campos Pimentel et al. (2019), Chen (2020), Muñoz-Carpio et al. (2020), Araiza-Alba et al. (2021), Daher and Sleem (2021), Huang et al. (2020), Guervós et al. (2019), Ding (2018), Granjo and Rasteiro (2018), Fong et al. (2018), Liu et al. (2019), Herault et al. (2018), Walk et al. (2020), Lin and Chen (2019), Malchow et al. (2018), Jung et al. (2018), Deb et al. (2017), Jo and Lim (2017), Nguyen and Liu (2016), Sharma et al. (2016a), Kohen-Vacs et al. (2016), Kleftodimos and Evangelidis (2016), Bao et al. (2019), To et al. (2016), Zhu et al. (2017), Llanda (2019), Liu et al. (2018), Lai et al. (2018), Singh et al. (2016), Khandwala and Guo (2018), Robal et al. (2018), Wachtler et al. (2016a), Koumaditis and Chinello (2019), Berns et al. (2018), Wachtler and Ebner (2019)	Quizzes embedded in the video, virtual reality and 360-degree characteristics, in-video annotations, feedback to the learner, games, hyperlinks, interface elements for navigation

Table 3 continued

Category	Sub-category	Examples
Production style	Cao et al. (2018), Ohashi et al. (2019), Hew and Lo (2020), Pi et al. (2017a), Stull et al. (2018), Srivastava et al. (2019), Davila et al. (2021), Srivastava et al. (2020), Lee and Muldner (2020), Nugraha et al. (2020), Pérez-Navarro et al. (2021), Al-Khateeb and Alduwairi (2020), Cookson et al. (2020), Ding et al. (2018), Stranc and Muldner (2019), Ding (2018), Costley and Lange (2017), Hildebrand and Ahn (2018), Rahim and Shamsudin (2019); Thornton et al. (2017), Shoufan (2019b), Lei et al. (2019), Shoufan and Mohamed (2017), Sharma et al. (2016b), Hartholt et al. (2020), Hayashi et al. (2020)	Khan-style, dialogue-style or monologue-style, slide-based, animations, talking head, paper-based explanations, classroom-style videos
Instructional design	Zhang (2021), van der Meij (2019), Eradze et al. (2020), Shoufan (2019b), Ou et al. (2016), Lei et al. (2019), Acuña-Soto et al. (2020), Zhang et al. (2018a), Minnes et al. (2019), Wijnker et al. (2019), Weinert et al. (2020), Ge and Li (2020), Lei et al. (2016), Voronkin (2019), Lange and Costley (2020), Stephenson (2019), Dodson et al. (2018b), Hodges (2018), Hung et al. (2018)	Multimedia learning principles, guidelines for effective engaging, according to film theory, interaction theory, limited capacity theory
Others	Schreiber and Dougherty (2017), Kikusawa et al. (2016), van der Meij and van der Meij (2016), der Zee et al. (2017), Shi et al. (2019)	Music, sign language, inclusion of a summary, visual complexity, content design: text layout, image position, text ratio, image ratio

characteristics of video-based learning materials (see examples on Table 3) such as in controlled experiments or tool development (this excludes, for instance, studies focusing on the feasibility of introducing videos in the classroom). These criteria ensure that our review specifically focuses on the evaluation and application of video characteristics that are relevant across diverse methodological research approaches, particularly those that inform learning effectiveness and their potential integration into AI-enhanced educational tools. Figure 1 shows the final set of articles, which encompasses 257 publications from 2016 to 2021, using the PRISMA guidelines.

In our data collection process, we noted the article's title, authors, year, source database, matching terms of our query set, methodological approach, research objectives, video characteristics, and, if applicable, main findings, discipline of the video, study sample size, video production style(s), dependent variables, and target tasks and technologies. All the information was assembled in a data table to facilitate efficient data access and analysis.

Literature Review

This section presents the results of the literature review. Based on our analysis, we distinguished four main methodological approaches: (1) methods, tools and frameworks ("tools" for conciseness), (2) controlled experiments, (3) data analyses, and (4) design guidelines. While broader analytical lenses could be applied, we selected these approaches because they offer distinct and complementary contributions to the understanding of video-based learning. Controlled experiments reveal trends in empirical validation and help identify which video features have a causal impact on learning effectiveness. Data analysis studies provide insights into learner behavior in authentic settings, capturing patterns that may not emerge in controlled environments. Tool development research bridges the gap between theory and practice by translating conceptual models into concrete applications. Finally, design guideline studies synthesize accumulated knowledge into actionable recommendations for educators and content creators. Together, these approaches allow us to examine both the empirical insights and practical implementations of VBL in a structured and meaningful way.

Of the reviewed articles, 57% presented tools and 25% described controlled experiments (see Fig. 2). If a study fitted multiple methodological approaches, we categorized it under the one that best reflected its primary focus and contributions. Figure 4

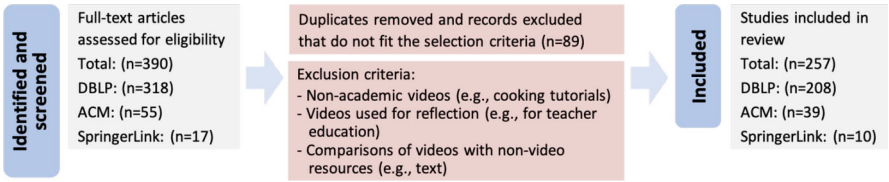

Fig. 1 The diagram shows the process of finding VBL-related articles in the databases. From an initial set of 390 papers, we removed duplicates and papers that did not fit the inclusion criteria. The final set consists of 257 articles from 2016 to 2021

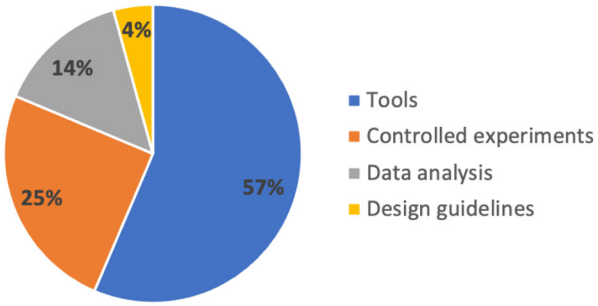

Fig. 2 The diagram shows the share of the different methodological approaches identified in this review. Tools and studies that perform controlled experiments constituted the main methodological approaches

depicts the structure of this section and the dimensions analyzed for each methodological approach. Table 3 summarizes the video characteristics we found and contextual aspects in a single taxonomy with eight categories: (1) audio features, (2) visual features, (3) textual features, (4) instructor behavior, (5) learner's activities, (6) interactive features, (7) production style, and (8) instructional design. Figure 3 shows the number of papers discussing a certain feature group, by the methodological approaches. Textual features were by far the most utilized features (Fig. 4).

Methods, Tools, and Frameworks

This section summarizes articles presenting tools to assist and enhance video-based learning as well as the analysis methods implemented to automatize tasks such as

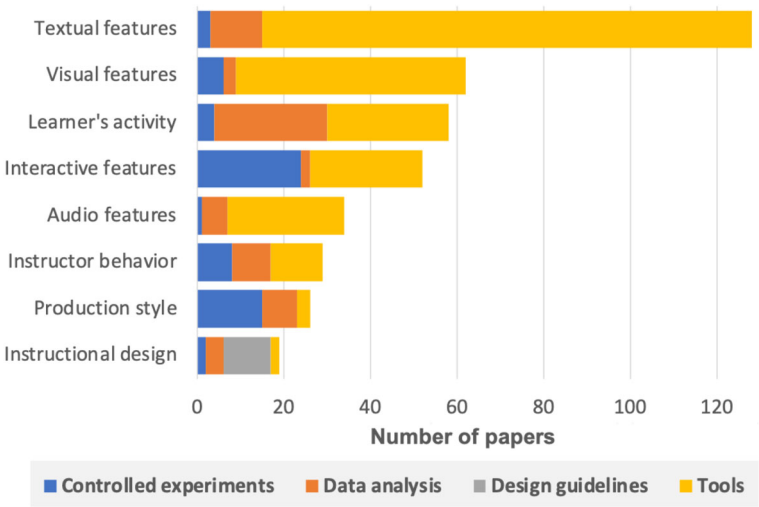

Fig. 3 The diagram shows the number of investigated video characteristics in the different methodological approaches. Some video characteristics were more widely explored in certain methodological approaches than in others

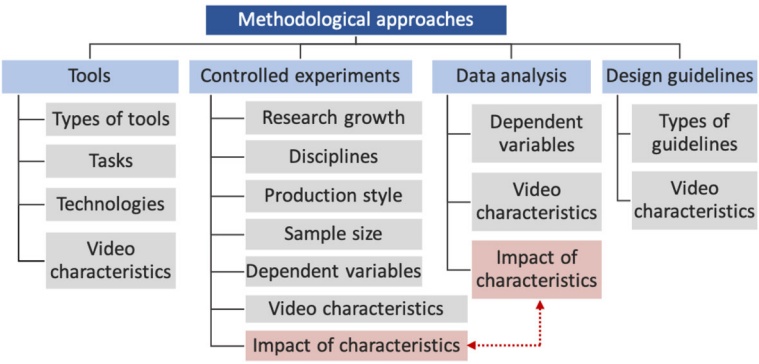

Fig. 4 The diagram shows the dimensions we analyzed for every methodological approach

information extraction. We discuss the following types of tools and methods: (a) information extraction methods, (b) interactive tools, (c) recommendation systems, (d) classifiers, (e) video navigation and segmentation tools, (f) video analysis tools and methods, (g) video summarization, and (h) others. The first five tool types amount to 80% of the reviewed papers (see Fig. 5). Some articles proposed systems consisting of two or more tools, e.g. Jung et al. (2018), Walk et al. (2020), Wachtler et al. (2016a), Khandwala and Guo (2018), and Zhang (2017).

Figure 6 shows the video characteristics used to develop different types of VBL tools. Text and visual features were most frequently used, mainly by recommender systems, information extraction and summarization methods, and navigational tools. Regarding tasks and technologies, we identified some low-level tasks performed as basic steps. For example, for textual video elements, several approaches employed optical character recognition (Xu et al., 2019b; Mahapatra et al., 2018a; Liu et al., 2018; Davila et al., 2021; Bian & Huang, 2019; Cagliero et al., 2019; Kota et al., 2019; Husain & Meena, 2019; Xu et al., 2019a; Davila & Zanibbi, 2018; Furini et al., 2018; Furini, 2018; Lee et al., 2017b; Waykar & Bharathi, 2017; Che et al., 2016; Balasubramanian et al., 2016; Dessì et al., 2017; Zhao et al., 2017; Jr. et al., 2019; Ponzanelli et al., 2019; Yadid & Yahav, 2016; Zhong & Ma, 2018; Mahapatra et al., 2018b; Nong et al., 2019; Khandwala & Guo, 2018), keyword extraction (Hernandez

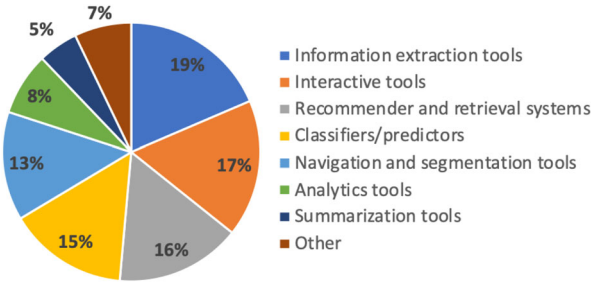

Fig. 5 The figure displays the types of tools that support and enhance video-based learning identified in this review

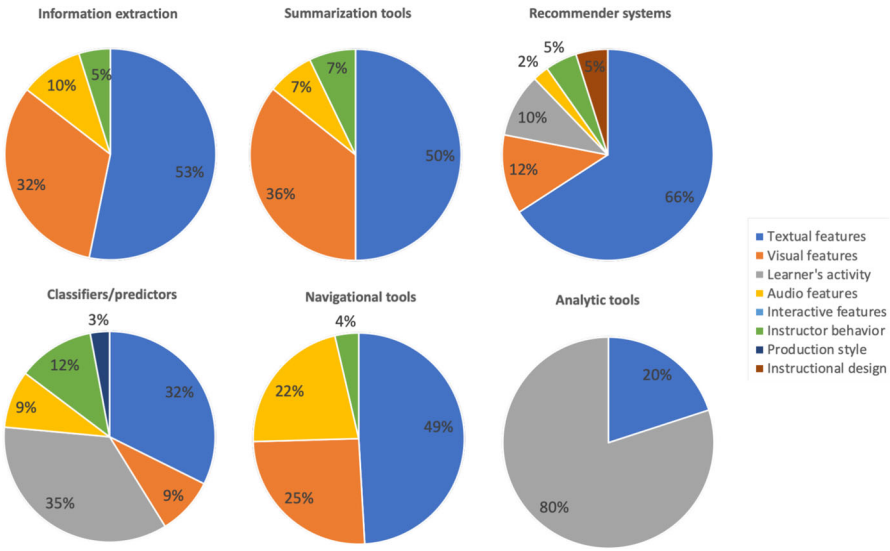

Fig. 6 The diagram displays the shares for video characteristics used to develop tools for VBL. Textual and visual features were the most used characteristics, mainly by recommendation systems, information extraction, summarization, and navigational tools

& Yang, 2021; Koka et al., 2020; Kravvaris & Kermanidis, 2019; Jo & Lim, 2017; Waykar & Bharathi, 2017; Kanadje et al., 2016; Balasubramanian et al., 2016; Zhang, 2021; Cooper et al., 2018; Coccoli & Vercelli, 2016; Kumar et al., 2017; Jr. et al., 2019; Escobar-Avila et al., 2017; Ji et al., 2018; Yadav et al., 2016; Mahapatra et al., 2018b; Schulten et al., 2020), generic natural language processing methods (e.g., Cagliero et al., 2019; Kravvaris & Kermanidis, 2019; Kravvaris & Kermanidis, 2017, van der Sluis et al., 2016; Poché et al., 2017), or utilized word embeddings (e.g., Ghauri et al., 2020; Hernandez & Yang, 2021; Nangi et al., 2019; Galanopoulos & Mezaris, 2019; Dessì et al., 2017; To et al., 2016). For image-related data, many suggested approaches for slide transition detection and video segmentation (Mahapatra et al., 2018a; Khattabi et al., 2019; Schroeder et al., 2020; Jung et al., 2018; Guan et al., 2019; Cagliero et al., 2019; Furini et al., 2018; Furini, 2018; Garber et al., 2017; Liu et al., 2017; Che et al., 2016; Nguyen & Liu, 2016; Zhao et al., 2017; Galanopoulos & Mezaris, 2019; Soares & Barrére, 2018; Basu et al., 2016a; Gandhi et al., 2016; Yadid & Yahav, 2016; Ghosh et al., 2021; Xu et al., 2019a). For audio tracks, various tools used speech recognition to generate transcripts (Leisner et al., 2020; Walk et al., 2020; Yoo et al., 2021; Bian & Huang, 2019; Ohnishi et al., 2019; Cagliero et al., 2019; Borges & dos Reis, 2019; Husain & Meena, 2019; Galanopoulos & Mezaris, 2019; Furini, 2018; Jo & Lim, 2017; Sanchez-Cortina et al., 2016; Balasubramanian et al., 2016; Basu et al., 2016b; Dessì et al., 2017; Coccoli & Vercelli, 2016; Gandhi et al., 2016; Zhao et al., 2017; Jr. et al., 2019; Mahapatra et al., 2018b; Chatbri et al., 2017). The next subsections discuss the tasks and technologies according to tool types, and the exploited video characteristics.

Information Extraction Methods

Information extraction approaches mainly focus on the extraction of text, discovery of additional metadata or important topics and segments in the video. These tasks have been considered in this review as they enhance the organization, accessibility, and contextualization of video materials- core tasks of AI-driven educational tools such as intelligent search, content summarization, and recommendation systems.

Extraction of text (handwriting, machine printed, and source code) Several approaches focus on the detection and recognition of handwriting: Davila et al. (2021) used a convolutional neural network (CNN); Kota et al. (Khattabi et al., 2019) used multiscale histogram of gradients for feature representation and deep neural networks; Kota et al. (2019) detected bounding boxes with handwriting and used clustering algorithms to group them. This approach could also identify content occluded by the instructor. Lee et al. (2017b) computed the increase in pixels detected as handwriting along consecutive video frames to identify the cutoff point when the text can be extracted before the instructor erases it. Time series denoising techniques are used for this purpose. Davila and Zanibbi (2018) extracted handwritten formulas using a random forest classifier. Some approaches identified instructor actions such as writing or erasing as clues to identify written video content (Xu et al., 2019a; Garber et al., 2017) via techniques such as pose estimation (Garber et al., 2017). Zhong and Ma (2018) extracted machine-printed text using the maximally stable extremal regions method to detect candidate regions and clustering techniques to ensemble lines of text. Khattabi et al. (2019) detected parts of the video that contained text using an AdaBoost meta-classifier. Finally, a few works focused on the extraction of source code (from coding lessons) (Khandwala & Guo, 2018; Yadid & Yahav, 2016; Alahmadi et al., 2018).

Extraction of key information Ghauri et al. (2020) identified key segments in videos using fusion schemes and bi-long-short-term memory network modules. Che et al. (2016) used the slides' structure and speech characteristics for the same task. Kravvaris and Kermanidis (2019) identified important topics in the transcripts using Latent Dirichlet Allocation, which were compared with relevant terms from the learners' comments. Lin et al. (2019) distinguished knowledge from non-knowledge segments using a CNN fed with image and sound simultaneously; however, it was not clarified the criteria to classify something as "knowledge". Similarly, Liu et al. (2018) extracted medical information according to the knowledge level (low or high) by building a Logistic Regression classifier using the text transcripts. For computer programming video tutorials, Ponzanelli et al. (2019) classified video segments according to the kind of content (concepts, software source code, etc.) The classifier employed temporal, structural, semantic, and code-related characteristics. Other works extracted key phrases from the videos (Yadid & Yahav, 2016; Albahr et al., 2021).

Extraction of metadata Jung et al. (2019) extracted keywords from transcripts using the TextRank algorithm. Also tags were extracted using singular value decomposition techniques, and named entities were extracted using entity recognition techniques. Escobar-Avila et al. (2017) compared term frequency-inverse document frequency, Latent Dirichlet Allocation using genetic algorithms, and the BM25F ranking func-

tion to find the best algorithms to extract tags from video content. Gomes et al. (Jr. et al., 2019) extracted related topics with the help of a knowledge base. A knowledge graph was built measuring the semantic similarity between concepts based on the number of resources the videos shared (Uniform Resource Identifiers of instances in the knowledge base) and the categories that the resources share. This was calculated with the Sorensen-Dice coefficient. Furini (2018) extracted tags by measuring the most frequent words. Ghosh et al. (2021) detected concepts that are unfamiliar or do not belong to the domain of the video but could be relevant to better understand the content. The concepts were linked to Wikipedia articles and used to build a concept similarity network. Unfamiliar concepts were extracted using community detection techniques. Similarly, Nangi et al. (2019) extracted unfamiliar topics also using community analysis in a concept graph.

Other works were dedicated to the extraction of slides (Guan et al., 2019; Liu et al., 2017).

Interactive Tools

Interactive tools allow for communication and interaction between the video and the learner. They include in-video annotations, in-video quizzes, and prevention of mind wandering.

In-video Annotations In most cases, note-taking tools were bundled with other supporting features, with only one exception that focused on annotation only (Deb et al., 2017). Fong et al. (2018) suggested a tool allowing learners to interact through annotations and in-video navigation. Liu et al. (2019) enabled the learner to create annotations and highlight transcript passages. Malchow et al. (2018) let the learner annotate the slides presented in the video. Lai et al. (2018) introduced text- and sketch-based annotations (by drawing into the video frame). Similarly, Singh et al. (2016) enabled text- and sketch-based annotations, with additional support tools for comment creation, color-coding, and navigation. Nguyen and Liu (2016) proposed a system that automatically extracts relevant parts of the content as candidate notes.

Quizzes Herault et al. (2018) proposed a 360-degree environment with embedded quizzes and feedback. Kohen-Vacs et al. (2016) introduced a tool to embed quizzes and announcements at any point in the video. Walk et al. (2020) included quizzes in the video to enhance active learning. Wachtler et al. (2016a) presented an application that includes single- and multiple-choice questions in the video. Ma and Ma (2019) developed an automatic question-generation method using the speech transcript and connecting to a knowledge graph. Wachtler and Ebner (2019) created an application to schedule quizzes and other interactive features in the video.

Feedback to learner for preventing mind wandering Lin and Chen (2019) tracked brainwave signals to detect loss of focus to recommend re-watching parts of the video. Jo and Lim (2017) used a gamified approach, presenting a word to the learner with the question if it was present in the video. Robal et al. (2018) used tracking (web activity,

mouse movement, and face) to determine whether the viewer was paying attention. If not, the system rang a bell, displayed a red frame around the video, and paused it. Sharma et al. (2016a) employed eye gaze tracking to identify mind wandering and displayed red marks when distraction was detected. Llanda (2019) tracked the users' emotions by analyzing face patterns and detecting negative sentiment to suggest re-watching the video if necessary.

Other Some studies proposed tools to create videos with 360-degree characteristics (Herault et al., 2018; Koumaditis & Chinello, 2019; Berns et al., 2018), clickable objects (Jung et al., 2018; To et al., 2016), interactive concept maps (Liu et al., 2018), feedback based on proficiency level (Zhu et al., 2017); or further interaction with the video content for practicing, for example, image editing (Kleftodimos & Evangelidis, 2016) and computer programming (Bao et al., 2019; Khandwala & Guo, 2018).

Recommender and Retrieval Systems

Recommender systems have mainly considered the similarity between transcripts and metadata to suggest new videos. Other approaches exploited, for instance, video popularity, learning style and teaching style.

Based on the similarity of transcripts and metadata Zhao et al. (2018a) suggested recommendations based on topic similarity, whereby the video is represented by means of its latent topics using Latent Dirichlet Allocation. Waykar and Bharathi (2017) retrieved videos based on keywords extracted from on-screen text and also the texture of keyframes using a texture histogram followed by Probability Extended Neighbor Classification to arrange the videos. Bleoanca et al. (2020) retrieved the top-N videos based on topic similarity using Latent Semantic Indexing to extract topics, clustering techniques to group the videos, and Wikipedia articles to augment the query context. Basu et al. (2016b) recommended videos based on the learner's navigation of Wikipedia pages, using Latent Dirichlet Allocation to extract topics from Wikipedia content. Jung et al. (2019) recommended similar videos based on keywords, tags, and readability features of the transcription coming from the instructor's speech. Similarly, Ji et al. (2018) generated recommendations using metrics of linguistic difficulty and similarity of keywords identified by the TextRank algorithm. Jordán et al. (2020) built a system that considers not only the characteristics of the current video but also previously watched videos. Walk et al. (2020) computed word embeddings of the video transcript and compared against the user search query to retrieve the most similar video segment. Tavakoli et al. (2020) built a video recommender system for learning skills required in the labor market; skills taught by a video are matched to the demands of job advertisements, combined with video popularity and length features.

Based on ontologies and external knowledge bases Schulten et al. (2020) first identify content concepts in the video and relate them to similar ones in DBpedia (Home, 2023). These are then used to discover other relevant learning resources through a web search API. In the same way, Borges and dos Reis (2019) extracted concepts and their relationships from an existing knowledge base to recommend videos. Coc-

coli and Vercelli (2016) linked the content to Wikipedia to extract concepts and their related concepts. The system then recommended both, videos and web pages for further learning. Borges and Silveira (2019) retrieved videos by analyzing the subject-object-predicate construction of the user query and matching them to a semantic graph database.

Playlists Aytekin et al. (2020) recommended sets of videos based on detected prerequisite relationships. Likewise, Tang et al. (2021) recommended a playlist by building a concept map that showed the relevant concepts in a set of videos and their relationships. The authors also considered the user sentiment derived from likes, dislikes, and comments. Furini et al. (2018) recommended a playlist composed of video segments covering a set of given keywords. They used silence periods and visual transitions to segment the videos relying on histogram-based techniques, and employed optical character recognition to extract video topics and keywords. Cooper et al. (2018) recommended videos based on the main topics extracted using the Latent Dirichlet Allocation algorithm and inter-topic relationships from the course syllabus. For creating the playlist, the authors used sequential pattern mining techniques to discover relationships.

Other Further recommender systems were based on video popularity (Kravvaris & Kermanidis, 2017; Zhang et al., 2018a), videos with the same instructor (Guan et al., 2019; Imran et al., 2019), viewer's learning style (Ciurez et al., 2019), the current software that the learner is using (Fraser et al., 2019), and multiple pedagogical criteria (Acuña-Soto et al., 2020).

Classifiers and Predictors

Classifiers and predictors were mainly used to estimate the expected learning outcome; some other noticeable tasks were discipline classification, instructor behavior prediction, and dropout prediction.

Learning outcome prediction Lemay and Doleck (2020) used a repeated incremental pruning algorithm to predict the learners' grades based on the frequency of video viewing. Several works aimed to predict whether a learner will pass or fail a course, mainly in MOOC contexts: Mbouzao et al. (2020) combined attendance (number of played videos divided by the number of mandatory videos), utilization (video play time divided by the sum of video lengths), and watching (attendance rate multiplied by utilization rate) metrics in a rule that compared each student against the average; Mubarak et al. (2021) trained a long-short-term memory network based on learner's activity data (e.g., play and pause); Furukawa et al. (2020) used the number of views, interval of video views (e.g., 0 to 19 days), and the first access to videos in a multiple regression predictor; finally, Korösi et al. (2018) used a random forest classifier and bagging based on characteristics related to platform usage (e.g., scrolling actions). In a different way to determine learning success, Lallé and Conati (2020) clustered learners into low and high achievers using characteristics such as the average proportion of videos watched per week. Aouifi et al. (2021), instead, predicted learning outcome

in terms of passing or failing the next quiz, based on previous quiz results fed into a recurrent neural network. Brinton et al. (2016) also predicted whether the learner will pass a quiz but using characteristics such as play, pause, revisiting, etc.

Discipline classification Stoica et al. (2019) and Dessì et al. (2017) used support vector machine algorithms to classify videos into their disciplines using textual features such as transcripts, keywords, and on-screen text. Basu et al. (2016a) clustered videos according to their disciplines by analyzing the topic similarity between the transcript and Wikipedia articles. Othman et al. (2017) classified videos into categories in academic scenarios such as Education and Science and Technology but also non-academic such as How-to and Style. The authors used decision trees and a Naive Bayes algorithm for classification based on metadata features. In a different approach, Chatbri et al. (2017) built a convolutional neural network fed with image-like representations (using statistical co-occurrence transformation techniques) that characterize the transcript and then classify the video into its discipline.

Instructor behavior prediction Chen et al. (2020) built a neural network to predict teachers' enthusiasm using facial expressions, voice, volume, and pose characteristics. Godavarthi et al. (2020) also used facial landmark characteristics but to predict instructor emotions using a CNN. Sharma et al. (2016b) predicted video liveliness (energetic or active) using two different long-short-term memory networks fed with audio features, visual features from the instructor actions, and production setup (e.g., instructor sitting or standing, slide or whiteboard usage). Xu et al. (2020) predicted the instructor's actions (write, pick eraser, erase, etc.) based on pose characteristics using a random forest and adaptive CNN classifiers.

Dropout prediction Jeon and Park (2020) used learners' activity data to predict weekly dropout in a MOOC by embedding click data sequences and feeding them into a gate recurrent unit. Furini et al. (2021) explored this task in an e-learning platform using features related to video access in random forest and K-nearest neighbor algorithms.

Other Further articles proposed models to predict how engaging a video was (Bulathwela et al., 2020), when a learner will jump back (Zhang, 2017), and to classify learners' comments (Poché et al., 2017).

Navigation and Segmentation Tools

Many articles proposed navigation in the style of a table of contents, but also navigation using keywords and visual elements has been presented. The suggested tools commonly used segmentation techniques to match the extracted information with the corresponding video segment.

Table of contents navigation Such tools consist of a table of contents which the user can click on and jump to the corresponding part in the video. They allow users to find segments that cover a certain sub-topic of interest. Indexing approaches allow to identify the video segments and the corresponding sub-topics or keywords (Das &

Das, 2020; Kravvaris & Kermanidis, 2019; Mahapatra et al., 2018a; Mukherjee et al., 2019; Husain & Meena, 2019; Gandhi et al., 2016; Kumar et al., 2017; Zhao et al., 2017; Yadav et al., 2016; Bian & Huang, 2019). Sub-topics and keywords have been extracted with technologies such as the Markov chain Monte Carlo method (Bian & Huang, 2019), Latent Dirichlet Allocation (Mukherjee et al., 2019; Kravvaris & Kermanidis, 2019; Husain & Meena, 2019; Bian & Huang, 2019), TextRank (Yadav et al., 2016; Mahapatra et al., 2018b), and concept extraction algorithms (Mahapatra et al., 2018b).

Keyword navigation These tools for navigation extracted keywords and indexed them according to specific segments of the video (Ohnishi et al., 2019; Koka et al., 2020; Kravvaris & Kermanidis, 2019; Kanadje et al., 2016; Kumar et al., 2017; Zhao et al., 2017).

Navigation using visual elements This kind of approach enabled exploration by presenting visual parts of the video, for example, main slides, figures, tables, charts, etc. The viewer was enabled to click and then jump to the corresponding segment in the video (Zhao et al., 2018b; Ma et al., 2017; Kumar et al., 2017; Zhao et al., 2017; Yadav et al., 2016).

Video segmentation When the production style is based on slides, one way to split the video was to detect slide transitions. For detecting slide transitions, edge-based differences between video frames (Zhao et al., 2017) and overlapping patches (Mahapatra et al., 2018a) were exploited. Other approaches proposed segmentation based on the video transcript. We observed TextTiling (Mukherjee et al., 2019; Zhao et al., 2017) for segmenting text, dynamic programming (Gandhi et al., 2016) to segment probable points where a topic changed, and the use of phrase extraction, named entity recognition, and part-of-speech-tagging technologies (Galanopoulos & Mezaris, 2019). Studies also used audio features to segment according to sound characteristics (e.g., power spectral density, fundamental frequencies, low-level acoustic characteristics) (Soares & Barrére, 2018, 2019).

Analytic Tools

Analytic tools relied heavily on the learner's activity data to understand video usage and its impact on learners' performance. A small sub-group of them used characteristics such as video metadata and eye gaze. The final results of analyses were often presented in the form of an informative dashboard.

Based on learner's activity Analytic tools often exploited log data from e-learning platforms to understand learner behavior. Walk et al. (2020) developed a tool that showed audience retention for video segments, the number of times the segment was re-watched, and how much time learners spent on it. Wachtler et al. (2016a)'s tool analyzed the video (e.g., number of views) and the student behavior (e.g., the extent to which the video was watched). Long et al. (2018) proposed a specific procedure

(analytics framework) to analyze patterns of usage in course videos, to understand, for instance, if all videos were watched, in their full length or only specific segments.

Analytic tools also used log data to understand the learner's behavior in relation to their performance. Min et al. (2019) suggested a system that contrasted the student performance on the quizzes with their watching behavior. He et al. (2018a) analyzed the viewing behavior by identifying if the learner watched the video and how much of the video was watched. Later, He et al. (2018b) showed how these metrics can be adopted to understand the results in academic performance.

Other analytic tool presented a method to embed the viewer's sequence of activities into a vector and analyzed if a video elicits particular behaviors (Mbouzao et al., 2019). Bothe and Meinel (2020) developed a metric to track viewers' visits to previous MOOC videos since learners do not always follow the path suggested by the platform. Choi et al. (2019) considered smartwatch data to improve the measurement of interest and difficulty.

Based on video metadata Othman et al. (2016) proposed a methodology to mine MOOC videos using their metadata (e.g., video length, comments, title) for several purposes such as clustering according to the discipline. Stepanek and Dorn (2017) presented a framework to analyze viewers' comments to identify and predict the engagement generated by a video. Sung et al. (2016) analyzed viewers' comments considering the sentiment, its relevance, its type, and its topic for displaying this information in the video timeline.

Other Zhang et al. (2018b) analyzed videos using data from eye-tracking systems.

Summarization Tools

Summarization approaches aim at generating a concise and shorter representation of a video. Text-based summarization was the most applied approach, however, summarization of visual elements was also suggested.

Text-based summarization These tools employ sentence segmentation, named entity recognition, extraction of key phrases and other metadata using transcripts (Vimalaksha et al., 2019; Balasubramanian et al., 2016; Kim & Kim, 2016; Yoo et al., 2021) and on-screen text (Xu et al., 2019b; Cagliero et al., 2019). Used supporting methods were term frequency-inverse document frequency (Vimalaksha et al., 2019), statistical analysis (Balasubramanian et al., 2016), Naive Bayes (Balasubramanian et al., 2016)), computation of cosine similarity between key concepts and the sentences in the video transcripts (Kim & Kim, 2016), OCR (Xu et al., 2019b), support vector machine (Xu et al., 2019b), and decision trees (Cagliero et al., 2019) algorithms to identify entities (Xu et al., 2019b), and Latent Semantic Analysis (Yoo et al., 2021).

Image-based summarization Rahman et al. (2020) proposed a visual summary by extracting a group of relevant figures (illustrations) from slide-based videos. The figure was included in the summary according to its importance, which was computed based on its size, unique aspects, and the duration of its occurrence in the video.

Other

Other articles proposed methods to create videos automatically (Hartholt et al., 2020; Hayashi et al., 2020; Furini, 2021), alleviate the split attention problem (Sharma et al., 2016c; Gopalakrishnan et al., 2017), enhance speech recognition (Hernandez & Yang, 2021; Sanchez-Cortina et al., 2016), and find similar content in other media such as textbooks (Wang et al., 2020a) and books (Horovitz & Ohayon, 2020).

Controlled Experiments

In this section, we report general findings regarding the evolution of experimental research, main disciplines, production style, and sample sizes. We also present the video characteristics that were tested in controlled experiments and the dependent variables used to measure the impact of those characteristics (e.g., on engagement, learning outcome, etc.).

Trends, Disciplines, Production Style, and Sample Size

Trends Articles describing controlled experiments became more frequent in the last three years in comparison to the three years prior. More than 60% of articles in this category were published in the last three years (Beege et al., 2020; van Alten et al., 2020; Leisner et al., 2020; Chen, 2020; Keller et al., 2019; Stranc & Muldner, 2019; Wang et al., 2020b; Schroeder et al., 2020; Moon & Ryu, 2021; Sözeri & Kert, 2021; Muñoz-Carpio et al., 2020; Cookson et al., 2020; Al-Khateeb & Alduwairi, 2020; Araiza-Alba et al., 2021; Taskin et al., 2019; Daher & Sleem, 2021; Pérez-Navarro et al., 2021; Lu et al., 2021; Nugraha et al., 2020; Wang et al., 2019a; Kuhail & Aqel, 2020; Huang et al., 2020; Garrett, 2021; Lee & Muldner, 2020; Lang et al., 2020; Chen et al., 2021; Srivastava et al., 2020; Horovitz & Mayer, 2021; Davila et al., 2021; Winkler et al., 2020; Pi et al., 2019; Srivastava et al., 2019; Zhang, 2021; Hoogerheide et al., 2018; Ku et al., 2019; Guervós et al., 2019; da Graça Campos Pimentel et al., 2019; Whitney & Dallas, 2019; Ohashi et al., 2019; van der Meij, 2019).

Disciplines Most of the experiments were performed using learning videos in STEM disciplines (around 80%) (Beege et al., 2020; Merkt et al., 2018; Beege et al., 2019; Kleftodimos & Evangelidis, 2018; Leisner et al., 2020; Keller et al., 2019; Ding et al., 2018; Schroeder et al., 2020; Moon & Ryu, 2021; Sözeri & Kert, 2021; Muñoz-Carpio et al., 2020; Al-Khateeb & Alduwairi, 2020; Pérez-Navarro et al., 2021; Lu et al., 2021; Nugraha et al., 2020; Garrett, 2021; Lee & Muldner, 2020; Lang et al., 2020; Chen et al., 2021; Srivastava et al., 2020; Horovitz & Mayer, 2021; Winkler et al., 2020; Pi et al., 2019; Srivastava et al., 2019; Stull et al., 2018; Wang et al., 2019b; Pi et al., 2017a, b; Schreiber & Dougherty, 2017; Li et al., 2016; Zhang, 2021; Sharma et al., 2016c; der Zee et al., 2017; Wachtler et al., 2016b; Cojean & Jamet, 2017; Ozdemir et al., 2016; Hoogerheide et al., 2018; Garrett, 2018; Ku et al., 2019; Palaigeorgiou & Papadopoulou, 2019; Hew & Lo, 2020; Guervós et al., 2019; van der Meij & van der Meij, 2016; Shelton et al., 2016; da Graça Campos Pimentel et al., 2019; Whitney &

Dallas, 2019; Ohashi et al., 2019; Cao et al., 2018; van der Meij, 2019). A few studies experimented with videos in both, STEM and non-STEM disciplines (e.g., Beege et al., 2020). Experiments with STEM topics have primarily considered physics, mathematics, and computer science subjects. In general, we identified the following STEM disciplines: (1) physics and mathematics (19% of controlled experiments) (Merkt et al., 2018; Leisner et al., 2020; Pérez-Navarro et al., 2021; Hoogerheide et al., 2018; Palaigeorgiou & Papadopoulou, 2019; Chen et al., 2021; Horovitz & Mayer, 2021; Wachtler et al., 2016b; Ku et al., 2019; Hew & Lo, 2020; Al-Khateeb & Alduwairi, 2020; Srivastava et al., 2020, 2019), (2) computer science (19%) (Sözeri & Kert, 2021; Muñoz-Carpio et al., 2020; Lee & Muldner, 2020; Lang et al., 2020; Winkler et al., 2020; Schreiber & Dougherty, 2017; Li et al., 2016; da Graça Campos Pimentel et al., 2019; Ohashi et al., 2019; Whitney & Dallas, 2019; Wang et al., 2019b; Nugraha et al., 2020; Chen et al., 2021), (3) biology, chemistry, and medicine (14%) (Ding et al., 2018; Zhang, 2021; Srivastava et al., 2020, 2019; Pi et al., 2019; Lu et al., 2021; der Zee et al., 2017; Stull et al., 2018; Beege et al., 2019; Guervós et al., 2019), (4) software use, digital media and communication (13%) (van der Meij, 2019; van der Meij & van der Meij, 2016; Garrett, 2021, 2018; Kleftodimos & Evangelidis, 2018; Keller et al., 2019; Pi et al., 2017a, b; Ozdemir et al., 2016), and (5) environmental science and meteorology (11%) (van der Meij, 2019; van der Meij & van der Meij, 2016; Garrett, 2021, 2018; Schroeder et al., 2020; Beege et al., 2020; Moon & Ryu, 2021; Sharma et al., 2016c; Cao et al., 2018; Cojean & Jamet, 2017; Shelton et al., 2016; Chen et al., 2021) Experiments on non-STEM topics focused mainly on language education (6%) (Chen, 2020; Kuhail & Aqel, 2020; Davila et al., 2021; Huang et al., 2020), history and geography (6%) (van Alten et al., 2020; Wang et al., 2020b; Daher & Sleem, 2021; Beege et al., 2020), and psychology (4%) (Cookson et al., 2020; Moos & Bonde, 2016; Chen et al., 2021).

Production style Most controlled experiments reported the production style of the videos (69%). We found that most of them used slide-based videos (39% of the experiments) (Leisner et al., 2020; Wang et al., 2019a; Lee & Muldner, 2020; Srivastava et al., 2020; Horovitz & Mayer, 2021; Pi et al., 2019; Srivastava et al., 2019; Wang et al., 2019b; Pi et al., 2017b; Sharma et al., 2016c; Hoogerheide et al., 2018; Kikusawa et al., 2016; Moos & Bonde, 2016; Hew & Lo, 2020; da Graça Campos Pimentel et al., 2019; Ohashi et al., 2019; Cao et al., 2018). A lot of experiments used videos with a visible instructor (36%) (Beege et al., 2020, 2019; Stranc & Muldner, 2019; Ding et al., 2018; Nugraha et al., 2020; Wang et al., 2019a; Srivastava et al., 2020; Horovitz & Mayer, 2021; Pi et al., 2019; Stull et al., 2018; Pi et al., 2017b; Hoogerheide et al., 2018; Kikusawa et al., 2016; Hew & Lo, 2020; da Graça Campos Pimentel et al., 2019; Cao et al., 2018). Less frequent production styles included handwriting (e.g., Khan-style, classroom-style videos) (16%), (Pérez-Navarro et al., 2021; Lee & Muldner, 2020; Srivastava et al., 2020; Stull et al., 2018; Schreiber & Dougherty, 2017; Sharma et al., 2016c; Hew & Lo, 2020), talking-head style (16%) (Pérez-Navarro et al., 2021; Davila et al., 2021; Pi et al., 2017a, b; Li et al., 2016; Hew & Lo, 2020; Cao et al., 2018), screencasts (14%) (Keller et al., 2019; Garrett, 2021; Ozdemir et al., 2016; Garrett, 2018; van der Meij & van der Meij, 2016; van der Meij, 2019), productions with virtual reality and 360-degree characteristics (14%) (Chen, 2020; Muñoz-Carpio et al.,

2020; Araiza-Alba et al., 2021; Daher & Sleem, 2021; Huang et al., 2020; Guervós et al., 2019), and animated videos (content created entirely with drawings, illustrations, or computer-generated effects) (14%) (Schroeder et al., 2020; Moon & Ryu, 2021; Cookson et al., 2020; Al-Khateeb & Alduwairi, 2020; Srivastava et al., 2020, 2019; Li et al., 2016). Interestingly, animations were considered more frequently in the last three years (Moon & Ryu, 2021; Cookson et al., 2020; Al-Khateeb & Alduwairi, 2020; Srivastava et al., 2020; Schroeder et al., 2020; Srivastava et al., 2019). Please note that some studies used more than one production style (e.g., Srivastava et al., 2020; Pi et al., 2017b; Hew & Lo, 2020).

Sample size The experiment with the least number of participants included four learners (Kikusawa et al., 2016), the one with most of participants included 398 learners (Shelton et al., 2016), the average was 94 participants. There was one experiment with a very large sample size of more than 12,000 participants, which was performed on a MOOC platform (der Zee et al., 2018).

Video Characteristics in Experiments

Figure 7 summarizes the video characteristics studied in controlled experiments. Specifically, we found four categories of characteristics that have been examined the most: (1) interactive features (e.g., quizzes) (39% of controlled experiments), (2) production style (e.g., talking head) (25%), (3) visual features (10%), and (4) instructor's behavior (e.g., gestures) (10%).

Interactive features These are features in the video that allow the learners to actively engage and interact with the content. The most used interactive features in experiments were quizzes embedded in the video (Kleftodimos & Evangelidis, 2018; Leisner et al., 2020; Wachtler et al., 2016b; Palaigeorgiou & Papadopoulou, 2019; da Graça

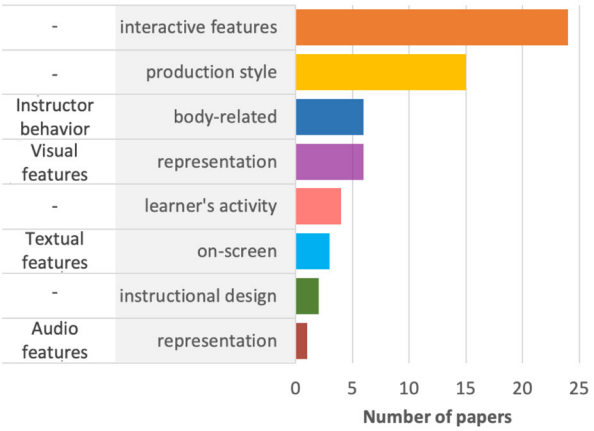

Fig. 7 The diagram shows the number of papers for each video feature type studied in controlled experiments. More than 80% of the studies focused on interactive features, production style, instructor behavior, and visual features

Campos Pimentel et al., 2019; Sözeri & Kert, 2021), virtual reality and 360-degree environments (Chen, 2020; Muñoz-Carpio et al., 2020; Araiza-Alba et al., 2021; Daher & Sleem, 2021; Huang et al., 2020; Guervós et al., 2019), in-video annotations (feature that allow the learner to take notes while watching the video) (Leisner et al., 2020; Taskin et al., 2019; Kuhail & Aqel, 2020; Chen et al., 2021), and navigational elements (e.g., table of content navigation, timelines with clickable points of interest) (Kuhail & Aqel, 2020; da Graça Campos Pimentel et al., 2019; Cojean & Jamet, 2017). Examples of less researched interactive features were games (e.g., crossword puzzles) (Cojean & Jamet, 2017), human-like conversational agents (Winkler et al., 2020), and self-regulated learning activities (van Alten et al., 2020; Moos & Bonde, 2016) (please refer to Section "Interactive Tools" for more information on technologies involved in the development of interactive features, and to Table 3 for the full list of researched interactive features).

Production style The production style refers to the scenery, the elements used to convey the information, and the physical arrangements considered to record the video; examples are Khan-style video, classroom-style videos, with talking head, etc. We use the typology of video production styles of Hansch et al. (2015). More than a third of studies that experimented with production styles addressed the instructor's presence (Cao et al., 2018; Hew & Lo, 2020; Pi et al., 2017a; Davila et al., 2021; Pérez-Navarro et al., 2021). Another third of the experiments evaluated dialogue style in contrast to the commonly used monologue style (Stranc & Muldner, 2019; Ding et al., 2018; Nugraha et al., 2020; Lee & Muldner, 2020; Ohashi et al., 2019). The last third of the experiments investigated animations (Cookson et al., 2020; Al-Khateeb & Alduwairi, 2020; Srivastava et al., 2020, 2019).

Visual and audio features This kind of experiments primarily addressed characteristics that attract the attention of the viewer, here called representation characteristics. The respective experiments mainly focused on directing attention using content-related cues (e.g., arrows, moving elements, color cues, gaze overlay) (Wang et al., 2020b; Moon & Ryu, 2021; Pi et al., 2019; Sharma et al., 2016c) and instructor-related elements (e.g., pointing gestures, spoken words, directed eye gaze) (Wang et al., 2020b; Moon & Ryu, 2021; Pi et al., 2019; Wang et al., 2019b; Pi et al., 2017b; Sharma et al., 2016c). Other experiments studied the impact of the complexity of visual content (Guervós et al., 2019) and the gender appearance of the instructor (Hoogerheide et al., 2018).

Instructor behavior characteristics In controlled experiments, the behavior of the instructors has been studied principally with respect to their gestures (Beege et al., 2020, 2019; Wang et al., 2019a; Horovitz & Mayer, 2021; Li et al., 2016). Less studied characteristics were related to the speech (e.g., tone of voice Horovitz & Mayer, 2021), other visual aspects such as pose (e.g., frontal, lateral), and dressing style (Beege et al., 2019).

Other characteristics Some articles studied the pace at which videos were viewed (Merkt et al., 2018; Schroeder et al., 2020; Garrett, 2021; Lang et al., 2020; Garrett,

2018). Among the less researched topics, we found experiments with captioned and non-captioned videos (Ozdemir et al., 2016; Whitney & Dallas, 2019), incorporating music (Schreiber & Dougherty, 2017), including summaries (van der Meij & van der Meij, 2016), including sign language (Kikusawa et al., 2016), and experiments that test Multimedia Learning Theory such as the effect of pre-training material and spatial contiguity between captions and visual elements (Zhang, 2021; van der Meij, 2019).

Dependent Variables

In controlled experiments, the dependent variable measured the impact of specific video characteristics on different learning aspects. We categorized the dependent variables into eight categories according to their purpose: (a) self-perceived effectiveness (e.g., self-efficacy, instructional efficiency) (Hew & Lo, 2020; Hoogerheide et al., 2018; Keller et al., 2019; Cojean & Jamet, 2017; van der Meij & van der Meij, 2016; da Graça Campos Pimentel et al., 2019; Schroeder et al., 2020; Moos & Bonde, 2016; Lee & Muldner, 2020; van der Meij, 2019; Huang et al., 2020; Sözeri & Kert, 2021), (b) difficulty (e.g., cognitive load, time effort) (Cao et al., 2018; Keller et al., 2019; Hoogerheide et al., 2018; Cojean & Jamet, 2017; da Graça Campos Pimentel et al., 2019; Beege et al., 2020; Lu et al., 2021; Srivastava et al., 2020, 2019; der Zee et al., 2017; Moos & Bonde, 2016; Merkt et al., 2018; Beege et al., 2019; Garrett, 2021; Lee & Muldner, 2020; Chen et al., 2021; Davila et al., 2021; Pi et al., 2017b; Moon & Ryu, 2021; Huang et al., 2020; Pi et al., 2017a), (c) general learning experience (e.g., lecture experience, learning experience) (Stull et al., 2018; Li et al., 2016; Palaigeorgiou & Papadopoulou, 2019; Shelton et al., 2016; Kleftodimos & Evangelidis, 2018; van Alten et al., 2020; Cookson et al., 2020; Pérez-Navarro et al., 2021; Lu et al., 2021; Wang et al., 2019a; Huang et al., 2020; Pi et al., 2017a; Cao et al., 2018; Chen, 2020), (d) emotions and feelings of the viewer (e.g., satisfaction, enjoyment) (Wang et al., 2019a; Horovitz & Mayer, 2021; Araiza-Alba et al., 2021; Hoogerheide et al., 2018; Srivastava et al., 2020, 2019; Schreiber & Dougherty, 2017; Stranc & Muldner, 2019; Cojean & Jamet, 2017; Guervós et al., 2019), (e) instructor-related characteristics (e.g., expressed enthusiasm, presentation skills) (Li et al., 2016; Hoogerheide et al., 2018; Al-Khateeb & Alduwairi, 2020; Hew & Lo, 2020; Cojean & Jamet, 2017; Lee & Muldner, 2020; Davila et al., 2021; Pi et al., 2017a; Wang et al., 2019b; Cao et al., 2018), (f) video and its content (e.g., quality, usability) (Guervós et al., 2019; Kikusawa et al., 2016; Palaigeorgiou & Papadopoulou, 2019; Ohashi et al., 2019; Keller et al., 2019; Daher & Sleem, 2021; Shelton et al., 2016; van der Meij & van der Meij, 2016) (g) engagement with the video (e.g., motivation, engagement) (Muñoz-Carpio et al., 2020; Cookson et al., 2020; Winkler et al., 2020; Shelton et al., 2016; Leisner et al., 2020; Araiza-Alba et al., 2021; Lee & Muldner, 2020; Srivastava et al., 2019; Wang et al., 2019b; Lu et al., 2021; Sharma et al., 2016c; Chen, 2020; Huang et al., 2020; Horovitz & Mayer, 2021; Ozdemir et al., 2016; Keller et al., 2019), (h) learning performance (e.g., learning quality, learning outcome) (Keller et al., 2019; Daher & Sleem, 2021; Cojean & Jamet, 2017; Merkt et al., 2018; Beege et al., 2019; Kleftodimos & Evangelidis, 2018; van Alten et al., 2020; Leisner et al., 2020; Chen, 2020; Ding et al., 2018; der Zee et al., 2018; Wang et al., 2020b; Schroeder et al., 2020; Moon

& Ryu, 2021; Sözeri & Kert, 2021; Cookson et al., 2020; Al-Khateeb & Alduwairi, 2020; Araiza-Alba et al., 2021; Taskin et al., 2019; Lu et al., 2021; Nugraha et al., 2020; Wang et al., 2019a; Kuhail & Aqel, 2020; Huang et al., 2020; Garrett, 2021; Lee & Muldner, 2020; Lang et al., 2020; Srivastava et al., 2020; Horovitz & Mayer, 2021; Davila et al., 2021; Pi et al., 2019; Stull et al., 2018; Wang et al., 2019b; Lin et al., 2018; Pi et al., 2017a, b; Schreiber & Dougherty, 2017; Li et al., 2016; Zhang, 2021; Sharma et al., 2016c; der Zee et al., 2017; Wachtler et al., 2016b; Cojean & Jamet, 2017; Ozdemir et al., 2016; Hoogerheide et al., 2018; Garrett, 2018; Ku et al., 2019; Palaigeorgiou & Papadopoulou, 2019; Moos & Bonde, 2016; Hew & Lo, 2020; van der Meij & van der Meij, 2016; Shelton et al., 2016; da Graça Campos Pimentel et al., 2019; Whitney & Dallas, 2019; Llanda, 2019; Cao et al., 2018; van der Meij, 2019), and (i) other (e.g., cybersickness, perceived similarity of the learner to the instructor) (Huang et al., 2020; Guervós et al., 2019; Cojean & Jamet, 2017; Beege et al., 2019; Ohashi et al., 2019; Hoogerheide et al., 2018; Hew & Lo, 2020).

Most of the controlled experiments examined some metric of learning outcome (86%), followed by cognitive load (20%), satisfaction (19%), mental effort (11%), social presence (learners' perception of their own social presence) (11%), motivation (9%), and self-efficacy (9%). Several studies examined more than one dependent variable.

The measurement of the learning outcome in several cases used metrics that evaluated *recall* (how much a learner can remember) (Kleftodimos & Evangelidis, 2018; Araiza-Alba et al., 2021; Taskin et al., 2019; Nugraha et al., 2020; Wang et al., 2019a; Davila et al., 2021; Li et al., 2016; Whitney & Dallas, 2019; Beege et al., 2020; Merkt et al., 2018; Beege et al., 2019; van Alten et al., 2020; Ding et al., 2018; Wang et al., 2020b; Schroeder et al., 2020; Lu et al., 2021; Kuhail & Aqel, 2020; Pi et al., 2019; Zhang, 2021; Ku et al., 2019; Hew & Lo, 2020) and *transfer* (the extent to which a learner can apply acquired knowledge to new contexts) (Garrett, 2018, 2021; Beege et al., 2020; Merkt et al., 2018; Beege et al., 2019; van Alten et al., 2020; Ding et al., 2018; Wang et al., 2020b; Schroeder et al., 2020; Lu et al., 2021; Kuhail & Aqel, 2020; Pi et al., 2019; Zhang, 2021; Ku et al., 2019; Hew & Lo, 2020). Transfer tests have been more frequently used in the recent four years than previously (Cojean & Jamet, 2017; Merkt et al., 2018; Beege et al., 2019; van Alten et al., 2020; Ding et al., 2018; Wang et al., 2020b; Schroeder et al., 2020; Lu et al., 2021; Kuhail & Aqel, 2020; Garrett, 2021; Pi et al., 2019; Zhang, 2021; Garrett, 2018; Ku et al., 2019; Hew & Lo, 2020). However, we found that only 38% of the articles on controlled experiments reported explicitly if the type of test was recall or transfer (Cojean & Jamet, 2017; Merkt et al., 2018; Beege et al., 2019; van Alten et al., 2020; Ding et al., 2018; Wang et al., 2020b; Schroeder et al., 2020; Lu et al., 2021; Kuhail & Aqel, 2020; Pi et al., 2019; Zhang, 2021; Ku et al., 2019; Hew & Lo, 2020; Kleftodimos & Evangelidis, 2018; Araiza-Alba et al., 2021; Nugraha et al., 2020; Davila et al., 2021; Li et al., 2016; Whitney & Dallas, 2019; Garrett, 2021, 2018). Furthermore, dependent variables were derived from tracking facial expressions (Srivastava et al., 2019), neurophysiological aspects (e.g., galvanic skin reaction) (Winkler et al., 2020), thermal aspects (facial temperature) (Srivastava et al., 2019, 2020), log data (da Graça Campos Pimentel et al., 2019; Lang et al., 2020), and eye gaze (Nugraha et al., 2020; Lee & Muldner, 2020; Stull et al., 2018; Wang et al., 2019b; Kikusawa et al., 2016; Pi

et al., 2019; Zhang, 2021; Sharma et al., 2016c; Moon & Ryu, 2021; Pi et al., 2017b; Srivastava et al., 2019).

Data Analysis Studies

The majority of data analysis studies focused on the understanding of patterns in learning behavior. Around half of the data analysis papers tried to understand the impact of behavioral patterns on some metric of learning effectiveness (Eradze et al., 2020; Stöhr et al., 2019; Wynn et al., 2019; Granjo & Rasteiro, 2018; Costley et al., 2021; Angrave et al., 2020; Li, 2019; Rodríguez et al., 2018; Costley & Lange, 2017; Brinton et al., 2016; Kovacs, 2016; Lee et al., 2017a; Hildebrand & Ahn, 2018; Shoufan, 2019a; Galster et al., 2018; Thornton et al., 2017; Atapattu & Falkner, 2017). Similar to controlled experiments, the impact of video characteristics has often been studied with focus on learning outcome (Costley & Lange, 2017; Stöhr et al., 2019; Wynn et al., 2019; Angrave et al., 2020; Shoufan, 2019a; Galster et al., 2018; Rodríguez et al., 2018; Brinton et al., 2016; Kovacs, 2016; Hildebrand & Ahn, 2018; Thornton et al., 2017; Li, 2019) and student engagement (Granjo & Rasteiro, 2018; Li, 2019; Costley & Lange, 2017; Galster et al., 2018; Thornton et al., 2017; Atapattu & Falkner, 2017; Hildebrand & Ahn, 2018).

Figure 8 sorts the characteristics studied in data analysis articles according to their frequency. We found that data analysis studies usually made use of learner's activity characteristics by taking advantage of system logs (e.g., play, pause) (Eradze et al., 2020; Saurabh & Gautam, 2019; Shoufan, 2019b; Stöhr et al., 2019; Wynn et al., 2019; Costley et al., 2021; Angrave et al., 2020; Li, 2019; Rodríguez et al., 2018; Costley & Lange, 2017; Giannakos et al., 2016; Hu et al., 2020; Uchidiuno et al., 2018; Kovacs, 2016; van der Sluis et al., 2016; Lei et al., 2019; Shoufan, 2019a; Meseguer-Martinez et al., 2019; Yan & Baxter, 2018; Atapattu & Falkner, 2017; Lee et al., 2017a; Lei et al.,

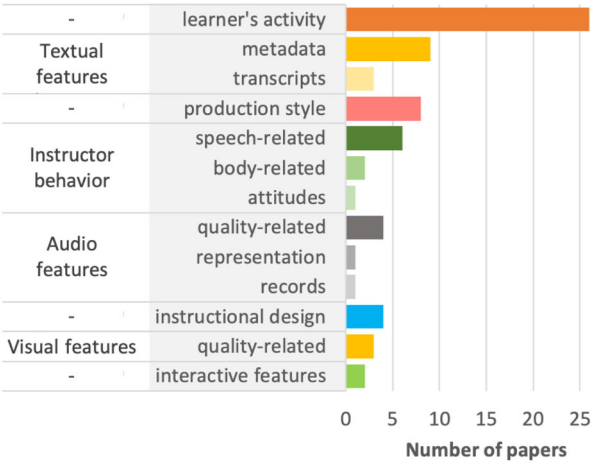

Fig. 8 The diagram displays the number of papers with regard to video characteristics explored in data analysis studies. These studies usually made use of log data of learners' activities

2017; Dodson et al., 2018a; Patel et al., 2018) and often analyzed metadata and video transcripts (Eradze et al., 2020; Saurabh & Gautam, 2019; Shoufan, 2019b; Ding, 2018; Shi et al., 2019; Lee et al., 2017a; Silva & Azevedo, 2017; Hildebrand & Ahn, 2018; Sjödén et al., 2018; van der Sluis et al., 2016; Meseguer-Martinez et al., 2019; Galster et al., 2018; Atapattu & Falkner, 2017). Moreover, these studies have inspected production style characteristics (Ding, 2018; Costley & Lange, 2017; Hildebrand & Ahn, 2018; Rahim & Shamsudin, 2019; Thornton et al., 2017; Shoufan, 2019b; Lei et al., 2019; Shoufan & Mohamed, 2017), audio features (Shi et al., 2019; Shoufan, 2019b; Ding, 2018; Shoufan & Mohamed, 2017) and visual features (Shoufan, 2019b; Ding, 2018; Shoufan & Mohamed, 2017), instructor's behavior characteristics (Ding, 2018; Tian & Bourguet, 2016; Shoufan, 2019b; van der Sluis et al., 2016; Atapattu & Falkner, 2017; Shoufan & Mohamed, 2017; Patel et al., 2018), design principles and guidelines (Eradze et al., 2020; Shoufan, 2019b; Ou et al., 2016; Lei et al., 2019), and interactive features (Ding, 2018; Granjo & Rasteiro, 2018). The findings are presented in the next section (Section "Impact of Video Characteristics on the Learning Effectiveness").

Design Guidelines

Guideline papers aim to derive recommendations on how to design learning videos to optimize learning processes. Common sources of insight were theoretical frameworks (Wijnker et al., 2019; Lange & Costley, 2020; Weinert et al., 2020), questionnaires to students (Wijnker et al., 2019; Lange & Costley, 2020; Dodson et al., 2018b; Minnes et al., 2019; Stephenson, 2019; Weinert et al., 2020), analysis of video content by experts (Wijnker et al., 2019; Lei et al., 2016; Ge & Li, 2020; Voronkin, 2019), interviews to teachers (Wijnker et al., 2019), assistants (Lei et al., 2016), and production staff (Lei et al., 2016), analysis of students' learning outcome (Minnes et al., 2019; Hung et al., 2018) and cognitive load (Hung et al., 2018), teacher experiences (Hodges, 2018; Stephenson, 2019), analysis of log data (Dodson et al., 2018b; Stephenson, 2019) and forum posts (Lei et al., 2016; Hung et al., 2018) of video platforms.

General guidelines Wijnker et al. (2019) developed a framework that applies a film theory perspective to learning videos. The authors provided guidance to analyze and categorize video according to their design, aim and type in film theory (Bordwell et al., 1993). Voronkin (2019) proposed universal guidelines focusing on methodological, psychological, didactic, ergonomic, and technical aspects. Lange and Costley (2020) considered media delivery problems and addressed them by proposing guidelines to improve pace, intelligibility, quality, media diversity, and congruence. The authors grounded their guidelines in the limited capacity theory in multimedia learning (Mayer, 2014), generative theory (Mayer, 2014), and extraneous processing research (Leppink et al., 2013). Dodson et al. (2018b) introduced a framework of active viewing that categorized the learners' watching behavior into interactive (e.g., collaborating, cooperating, communicating), constructive (e.g., taking notes), and active behavior (e.g., browsing, searching, pausing). The authors stated that VBL can be designed in a way to promote this behavior.

Guidelines for specific disciplines Minnes et al. (2019) proposed design guidelines for producing videos for computer science and endorsed certain representation characteristics, especially call-out boxes, zooming, and animations. Hodges (2018) also focused on computer science and suggested to use captions, descriptions of animations as they occur, zooming in relevant code, and systems that allow to pause, seek, and speed up or slow down the video. Stephenson (2019) concentrated on the development of programming videos following a specific content structure. Lei et al. (2016) discussed an approach to produce videos for Humanities disciplines. They suggested favoring short videos, using animations, highlighting key concepts and terminologies, including an introduction and conclusions, employing consistent topography and graphic styles, and adding music in specific parts. Ge and Li (2020) identified beneficial characteristics to learn English as a foreign language. The recommendations include attention to speech rate, sentence, and vocabulary complexity, use of closely related examples, use of realistic visuals when higher levels of context are demanded, including a visible instructor, and use of keyword subtitles instead of complete subtitles.

Guidelines for videos with interactive features Weinert et al. (2020) introduced guidelines with reference to Design Interaction Theory (Moore, 1989), Cognitive Load Theory (Sweller, 1994), and own research. They recommended to not only include elements for interaction with the content but also with other learners and the instructor. They further discussed recommendations for positioning specific interactive features. Hung et al. (2018) proposed guidelines to design interactive tasks in learning videos based on the literature on embodied cognition. The authors propose that the video should include engaging, prompting, experiencing, facilitating, demonstrating, and questioning tasks.

Impact of Video Characteristics on the Learning Effectiveness

In this section, we consolidate and summarize the findings presented in articles on controlled experiments and data analyses. They are organized according to the categories of video characteristics as presented in Table 3.

Instructor Behavior Characteristics

Gestures Experiments showed that gestures do not have a significant impact on the learning outcome (Beege et al., 2020; Wang et al., 2019a; Horovitz & Mayer, 2021). However, learners still cared about gestures, facial expressions, and body language (Shoufan, 2019b; Ding, 2018). Among the benefits, gestures reduced mental effort (Beege et al., 2020), increased presence (Beege et al., 2020; Li et al., 2016) and satisfaction (Wang et al., 2019a). Moreover, studies obtained a positive effect on delayed learning outcome tests and a higher impact of gestures from human instructors compared to animated instructors (human-like artificial agents) (Wang et al., 2019a; Horovitz & Mayer, 2021).

Speech Data analysis studies revealed that speech characteristics such as speaking rate, lexical diversity, etc., affected how learners interacted with the video (Atapattu & Falkner, 2017). Speech characteristics were also found to correlate with sound characteristics (Shi et al., 2019). Further findings showed that they impacted dwell time (van der Sluis et al., 2016) and the number of likes and dislikes a video can obtain (Shoufan, 2019b; Shoufan & Mohamed, 2017). For example, whether the instructor was a native speaker or not significantly affected the number of likes and dislikes (Shoufan, 2019b; Shoufan & Mohamed, 2017). Also, pronunciation and intonation affected the learner's attention (Shoufan, 2019b). Based on these discoveries, it has been recommended to reduce lexical diversity, long sentences, and speech rate (Atapattu & Falkner, 2017).

Other Other characteristics related to instructor behavior that impacted learners were the appearance, clothes, and enthusiasm (Shoufan, 2019b; Ding, 2018). Interestingly, gender did not have a significant impact (Shoufan, 2019b), it did not affect the learning outcome, the perception of self-efficacy (Hoogerheide et al., 2018), and learners liking or disliking a video (Shoufan & Mohamed, 2017). Yet, it was observed that learners perceived themselves as more similar to an instructor of their own gender (Hoogerheide et al., 2018). With regard to animations as instructors, these were considered interesting and attractive (Ding, 2018).

Interactive Features

Quizzes Most of the studies experimenting with quizzes evidenced a positive effect on learning outcome (Kleftodimos & Evangelidis, 2018; Leisner et al., 2020; Wachtler et al., 2016b; Palaigeorgiou & Papadopoulou, 2019; da Graça Campos Pimentel et al., 2019). Palaigeorgiou and Papadopoulou (2019) found a positive effect when quizzes were combined with other interactive features; Wachtler et al. (2016b) when they were placed at the right time. Further positive impacts of quizzes were observed on aspects as satisfaction (Kleftodimos & Evangelidis, 2018; Palaigeorgiou & Papadopoulou, 2019) and engagement (da Graça Campos Pimentel et al., 2019). Data analysis studies also confirmed these findings: Kovacs (2016) found that most viewers engaged in quizzes, which seemed to reduce dropout. Ding (2018) showed that quizzes improved understanding, but did not draw the learner's attention when too complex or placed at the end of a video.

Virtual reality and 360-degree characteristics Articles in these categories commonly reported an improved learning outcome, however, not to a significant degree (Chen, 2020; Araiza-Alba et al., 2021; Daher & Sleem, 2021; Huang et al., 2020). Huang et al. (2020) found positive effects only in certain parts of the post test. Although statistically insignificant, Araiza-Alba et al. (2021) reported short- and long-term improvements in learning outcome. Studies also showed differing impacts of virtual reality and 360-degree environments on learner motivation: While Huang et al. (2020) found no positive effect on the motivation, Muñoz-Carpio et al. (2020) and Chen (2020) found that viewers felt more motivated. Regarding satisfaction, several

findings confirmed an increase (Chen, 2020; Muñoz-Carpio et al., 2020; Huang et al., 2020). Furthermore, Huang et al. (2020) found a positive impact on self-efficacy. Our analysis over time shows that these characteristics were investigated more frequently in recent years (Chen, 2020; Muñoz-Carpio et al., 2020; Huang et al., 2020; Araiza-Alba et al., 2021; Daher & Sleem, 2021). Naturally, these experiments evaluated the impact in other non-common dependent variables such as sense of "being there" (Araiza-Alba et al., 2021; Guervós et al., 2019) and cybersickness (Guervós et al., 2019). Interestingly, most of them were conducted with non-STEM videos (Chen, 2020; Araiza-Alba et al., 2021; Daher & Sleem, 2021; Huang et al., 2020).

Other Studies that experimented with in-video annotations found an increase in the learning outcome (Leisner et al., 2020; Taskin et al., 2019; Kuhail & Aqel, 2020), whereas neutral effects were found for cognitive load (Chen et al., 2021). Regarding navigational features, experiments using table content navigation (Kuhail & Aqel, 2020; Cojean & Jamet, 2017) and timelines with clickable points (da Graça Campos Pimentel et al., 2019; Cojean & Jamet, 2017) generally found a positive effect on the learning outcome. However, this was not always significant (da Graça Campos Pimentel et al., 2019), or was examined combined with other features (e.g, table contents together with timelines with markers (Cojean & Jamet, 2017)). It was suggested that videos with interactive features need to be used with other pedagogical techniques to improve the learning outcome (Granjo & Rasteiro, 2018).

Visual and Audio Features

Instructor-related representational characteristics Tian and Bourguet (2016) observed that the most common gestures that instructors used as cues were pointing and extending (extending the arms away from the body with palms up). The results showed that these cues increase the learning outcome in most cases (Wang et al., 2020b; Pi et al., 2019; Wang et al., 2019b; Pi et al., 2017b). However, no significant effects (Sharma et al., 2016c) or even negative effects were found when the instructor was an animation (Moon & Ryu, 2021).

Content-related representational characteristics Examples of these characteristics are arrows, color-coded cues, and highlighted content. Experiments showed that content-related cues improved the learning outcome in most cases (Wang et al., 2020b; Pi et al., 2017b; Sharma et al., 2016c), and had no significant effect on cognitive load (Sharma et al., 2016c). Comparing content-related and instructor-related cues, the results were diverse: In one case, instructor-related cues had a higher positive impact on the learning outcome (hand-pointing gestures vs. arrows on slide-based videos) (Pi et al., 2017b). Contrary results were found when comparing a pencil following the instructor's handwriting (instructor-related cue) with a highlighted "optimal gaze path" defined by an expert (visual cue). The recommended gaze direction yielded a more positive impact (Sharma et al., 2016c).

Other visual and audio elements Visual and audio features, such as image and audio quality, mattered to the learners and caused positive effects on the learning outcome (Ding, 2018; Schreiber & Dougherty, 2017). Uchidiuno et al. (2018) showed that students who are not native English speakers but are learning the language often avoided videos with no visual support (figures, charts, equations, or on-screen text). Ding (2018) found that background music helped students to stay focused.

Production Style

Instructor presence Hew and Lo (2020) sampled a set of computer science-themed YouTube videos where it can be observed that the three most widely used production styles showed an instructor: (1) slide-based videos, (2) classroom-style videos, and (3) Khan-style videos with a talking head. In addition, viewers voiced their preference for videos that showed the instructor's hands (Pérez-Navarro et al., 2021). However, although the presence of an instructor was preferred, it did not bring notable advantages: Two studies found that a talking head did not show a significant effect on the learning outcome (Davila et al., 2021; Hew & Lo, 2020), social presence (Hew & Lo, 2020) and cognitive load (Davila et al., 2021). Experiments also tested the effects of the instructor's size in the video. A small size seemed to cause higher learning gains (Pi et al., 2017a) and satisfaction when the topic was not complex (Cao et al., 2018). However, it had no significant effect on cognitive load (Pi et al., 2017a), mental effort (Cao et al., 2018) and social presence (Pi et al., 2017a; Cao et al., 2018).

Dialogue-style vs. Monologue-style The experiments did not observe a significant effect of dialogue-style over monologue-style videos on learning gain (Ding et al., 2018; Nugraha et al., 2020; Lee & Muldner, 2020). In addition, no effect on social presence was found (Lee & Muldner, 2020). Ohashi et al. (2019) conducted an experiment using videos on computer programming, one using slides only, and another one accompanying the slides by two puppets in a dialogue. They found that the number of viewers of the video including the dialogue decreased over time (throughout a compulsory course). The authors argued that the puppets' dialogue was effective at the beginning, but not when the learners became increasingly familiar with the course content. Nevertheless, the dialogue style seemed to offer benefits with regard to self-efficacy (Lee & Muldner, 2020) and positive sentiment (Ding et al., 2018).

Animations Animations in videos caused higher learning outcomes when compared to (a) slide-based videos only with text and without illustrations, figures, and similar (Srivastava et al., 2020), (b) textbooks (Cookson et al., 2020), and (c) live classroom lectures (Al-Khateeb & Alduwairi, 2020). They also had a positive impact on aspects such as participation (Cookson et al., 2020), cognitive load (Lee & Muldner, 2020), mental effort (Srivastava et al., 2020), and perceived difficulty (Srivastava et al., 2019).

Video tutorials and video lectures Learners stated that video tutorials were more interesting, and attractive, kept them engaged, and helped them understand better (Ding,

2018; Ou et al., 2016). One study found that learners tend to skip theoretical discussions, usually found in video lectures (Lei et al., 2019). However, other studies showed that learners not only watched lectures more often than tutorials (Hildebrand & Ahn, 2018) but also watched them more often to completion (Thornton et al., 2017). Hildebrand and Ahn (2018) reported that both styles had a similar positive correlation with the learning outcome.

Other Shoufan and Mohamed (2017) reported that the most predominant styles in CS videos were paper-based explanations, whiteboard-based explanations, slide-based videos, and Khan-style videos. Regardless of specific styles one study found that, in general, media diversity correlated positively with learning outcome, satisfaction, engagement, and interest (Costley & Lange, 2017).

Pace of the Video

Automatic pauses vs. non-automatic pauses No significant positive effects of automatic pauses in videos were found compared to non-automatic pauses (where learners can pause the video, also called self-pacing) (Schroeder et al., 2020). Indeed, systems with automatic pauses in videos were ineffective (Garrett, 2021) or even increased cognitive load (Garrett, 2021, 2018). Merkt et al. (2018) did not find advantages of automatic pauses even in long lasting videos. On the contrary, viewers in the non-automatic pauses condition were more successful and reported a lower mental effort (Garrett, 2018; Schroeder et al., 2020).

Video acceleration The video acceleration was found to be important for learners (Ding, 2018). There was evidence that students tend to accelerate the videos at least once (Yan & Baxter, 2018). Lang et al. (2020) showed that when the learner accelerated the video, the learning outcome improved for learners on Stanford Online (veena, 2020) (a platform of open online courses) in diverse disciplines. Garrett (2021) reported that learners who watched accelerated CS videos and had the control over playing and rewinding outperformed learners who also watched accelerated videos but did not have control over these features.

Beginners, Specialists, High, and Low Performers

Studies observed specific behaviors in learners, depending on their performance level with respect to a learning topic. High achievers engaged in more diverse activity patterns (e.g., play, pause seek) (Rodríguez et al., 2018), interacted more intensely with embedded quizzes (Kovacs, 2016), and watched videos more consistently (Rodríguez et al., 2018). Galster et al. (2018) reported that students who were more active through in-video annotations also watched more videos. However, differences in learning outcome between more engaged and less engaged learners were not found (Galster et al., 2018). One study suggested that low-performance learners could gain advantages by imitating high achievers behavior (Angrave et al., 2020).

Regarding the differences between beginners and specialists, one study showed that specialists engaged more often in different video activities than beginners, although both consumed the same amount of videos (Li, 2019). Contrarily, another study did not find a difference in the level of engagement in comparison to beginners; however, specialists achieved a higher learning outcome (Stöhr et al., 2019).

Video Completion and Length

Video completion It was reported that learners usually watched 30% to 50% of the video (on YouTube Saurabh & Gautam, 2019 and on the MOOC platform edX Stöhr et al., 2019; edX, 2025) and that skipping through videos was a very common behavior (Hildebrand & Ahn, 2018). However, studies in other learning contexts obtained different results. If the videos were part of a university course, most of the learners watched the full video (Yan & Baxter, 2018; Giannakos et al., 2016). Regarding the learning outcome, studies showed that the extent to which a video was watched impacted the students' grades (Hildebrand & Ahn, 2018; Angrave et al., 2020) and that viewers who watched the video to completion were more likely to watch other videos (Wynn et al., 2019). Dropout usually happened within the first five seconds (Wynn et al., 2019). Once the learners watched the video for a longer time, dropout rates decreased (Wynn et al., 2019). Wynn et al. (2019) found that after watching the first five minutes, most viewers completed a video. Strategies such as including in-video quizzes seemed to reduce dropout (Kovacs, 2016).

Video length Several studies have concluded that short learning videos were preferred by learners (Saurabh & Gautam, 2019; Ding, 2018; Engeness & Nohr, 2019). Videos within five to seven minutes were preferred in two studies (Engeness & Nohr, 2019; Ding, 2018). One of the studies made this finding by analyzing the comments of students written down after watching the educational videos. This study experimented with videos with talking head, slides, drawings on a board, and animations for learning English (Ding, 2018). The other study surveyed students concerning their preferred length of MOOC videos. This study used MOOC videos on digital competence that presented pedagogical aspects and tutorials for teachers (Engeness & Nohr, 2019). Further findings in both studies showed that although longer videos up to 10 minutes were not preferred, they were still well received (Engeness & Nohr, 2019; Ding, 2018). Saurabh and Gautam (2019) found that the most popular learning videos lasted 10 to 15 minutes. It was, however, also shown that videos should not be too short to be perceived positively by the learners (Shoufan, 2019a). The authors compared YouTube videos with different liking ratios (number of likes of a video divided by the sum of likes and dislikes), different instructors, different speaking accents, and covering engineering topics such as digital logic design and embedded systems. For lecture styles, Giannakos et al. (2016) surveyed students to understand their attitudes and usage patterns. They found that the preferred length of videos in software engineering was close to the traditional 45-minute classroom lectures for university students of a computer science degree. In any case, the context was an important factor to consider when selecting an appropriate video length (Shoufan, 2019a).

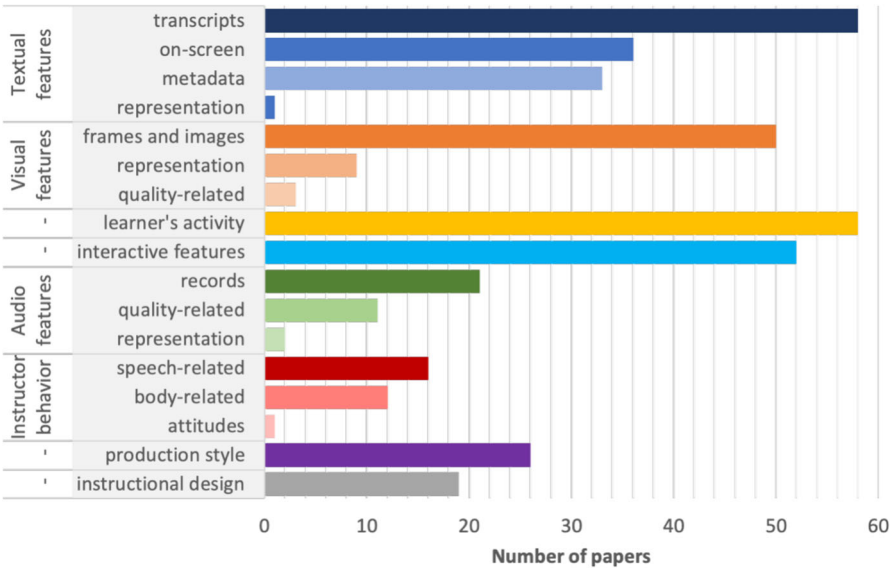

Fig. 9 The diagram shows all video characteristics identified in this review along with the number of papers where they were present

Video Popularity

Saurabh and Gautam (2019) found that viewers tend to like videos that helped them to learn and dislike the ones that did not. Shoufan and Mohamed (2017) discovered that the number of likes for learning videos increase (a) with higher video resolution, (b) with higher talking rates (words per second), (c) when English was the native language of the instructor, (d) when the production style included paper-based or whiteboard-based explanations, and (e) when more than one production style was used. Later, Shoufan (2019a), found in a survey that learners liked a video when they perceived that (a) the explanation seemed appropriate, (b) the presentation used animations, diverse colors, legible handwriting, and appropriate supporting software, (c) the video seemed interesting and easy to focus on, (d) practical examples were included, and when (e) the instructor was concise, used a simple vocabulary, and had a confident and clear voice. On the other hand, in a different study, Shoufan (2019b) found that learners perceived that they disliked a video when they felt like they did not understand the video content, when the technical quality of audio or video was insufficient, when the instructor seemed incompetent, not friendly, or not enthusiastic, or the length of the video was not appropriate. Curiously, it was found that lengthy videos generated slightly fewer dislikes (Shoufan & Mohamed, 2017).

Lee et al. (2017a) detected that popular videos have a high amount of comments with strong negative sentiment. Moreover, Lei et al. (2019) found that popular videos had a high number of views but lower retention rates. The authors found that popular videos usually provided a general background, while less popular videos dived deeper into the topic, which might be the reason why they had higher retention rates. Indeed,

it was warned that a higher amount of views and comments might not necessarily entail high video quality (Lee et al., 2017a).

Conclusions

In this article, we have presented a review of 257 articles from the field of video-based learning (2016-2021). We (a) identified main methodological approaches (b) identified video characteristics and organized them into a taxonomy, (c) reported on technologies, (d) revealed trends in design guidelines to produce videos, and (e) summarized the impact of the features on the learning effectiveness.

Methodological approaches We identified four approaches in VBL: (1) proposals of tools, (2) reports on controlled experiments, (3) data analysis studies, and (4) design guidelines proposals. Most of the papers suggested tools for supporting VBL, a moderate part were controlled experiments, and a small part was data analysis and design guidelines articles.

Video characteristics Table 3 presents the taxonomy we built based on the characteristics investigated in the reviewed articles. Figure 9 ranks these video characteristics according to the number of related publications. The **most explored characteristics** were textual features followed by visual, learner activity, and interactive features. The most used textual features were transcripts, mainly employed in recommendation systems (Section "Recommender and Retrieval Systems"), navigation tools (Section "Navigation and Segmentation Tools"), information extraction methods (Section "Information Extraction Methods"), and summarization tools (Section "Summarization Tools" and Fig. 6). The most explored visual features were video frames and information extracted from them. Learner activity characteristics took advantage of log data (e.g., play and pause). These were heavily explored in data analysis articles (Section "Data Analysis Studies" and Fig. 3) to understand learner behavior. Interactive features were popular in experiments to evaluate their effectivity (Kleftodimos & Evangelidis, 2018; van Alten et al., 2020; Leisner et al., 2020; Keller et al., 2019; der Zee et al., 2018; Sözeri & Kert, 2021; Taskin et al., 2019; Lu et al., 2021; Kuhail & Aqel, 2020; Chen et al., 2021; Winkler et al., 2020; Wachtler et al., 2016b; Cojean & Jamet, 2017; Ku et al., 2019; Palaigeorgiou & Papadopoulou, 2019; Shelton et al., 2016; Moos & Bonde, 2016; da Graça Campos Pimentel et al., 2019; Chen, 2020; Muñoz-Carpio et al., 2020; Araiza-Alba et al., 2021; Daher & Sleem, 2021; Huang et al., 2020; Guervós et al., 2019). **Moderately studied characteristics** included audio features, production style, and instructor behavior. Audio features were often used to develop tools (Mahapatra et al., 2018a; Ohnishi et al., 2019; Husain & Meena, 2019; Soares & Barrére, 2018; Kanadje et al., 2016; Gandhi et al., 2016; Kumar et al., 2017; Zhao et al., 2017; Mahapatra et al., 2018b; Bian & Huang, 2019; Balasubramanian et al., 2016; Walk et al., 2020; Furini, 2018; Jr. et al., 2019; Lin et al., 2019; Sharma et al., 2016b; Chatbri et al., 2017; Furini, 2021; Sanchez-Cortina et al., 2016; Gopalakrishnan et al., 2017; Mukherjee et al., 2019; Soares & Barrére, 2018; Kanadje et al., 2016; Ghauri et al., 2020; Furini, 2018; Imran et al., 2019; Chen et al., 2020). Production styles were evaluated for their effectivity (Stranc & Muld-

ner, 2019; Ding et al., 2018; Cookson et al., 2020; Al-Khateeb & Alduwairi, 2020; Pérez-Navarro et al., 2021; Nugraha et al., 2020; Lee & Muldner, 2020; Srivastava et al., 2020; Davila et al., 2021; Srivastava et al., 2019; Stull et al., 2018; Pi et al., 2017a; Hew & Lo, 2020; Ohashi et al., 2019; Cao et al., 2018) (Fig. 3). Regarding instructor behavior, speech-related characteristics (Horovitz & Mayer, 2021; Shoufan, 2019b; Ding, 2018; van der Sluis et al., 2016; Atapattu & Falkner, 2017; Shoufan & Mohamed, 2017; Patel et al., 2018; Mahapatra et al., 2018a; Soares & Barrére, 2019; Balasubramanian et al., 2016; Jo et al., 2019; Jung et al., 2019; Che et al., 2016; Jung et al., 2019; Ji et al., 2018; Bulathwela et al., 2020) have been explored more often than body-related characteristics (Beege et al., 2020, 2019; Wang et al., 2019a; Horovitz & Mayer, 2021; Li et al., 2016; Ding, 2018; Tian & Bourguet, 2016; Chen et al., 2020; Xu et al., 2020; Godavarthi et al., 2020; Hoogerheide et al., 2018; Shoufan & Mohamed, 2017) (Fig. 9).

Tools, tasks and technologies The main tools developed for VBL support were information extraction, integration of interactive features, recommendation systems, classification and prediction, video navigation, and summarization (Section "Methods, Tools, and Frameworks" and Fig. 5). Table 4 provides details for each specific category along with its implementation frequency.

Design guidelines Some studies proposed general guidelines (Wijnker et al., 2019; Voronkin, 2019; Lange & Costley, 2020; Dodson et al., 2018b), others suggested guidelines to produce videos in specific disciplines (Minnes et al., 2019; Hodges, 2018; Stephenson, 2019; Lei et al., 2016; Ge & Li, 2020), and two focused on interactive videos (Weinert et al., 2020; Hung et al., 2018). They leveraged theoretical frameworks (Wijnker et al., 2019; Lange & Costley, 2020; Weinert et al., 2020), questionnaires to students (Wijnker et al., 2019; Lange & Costley, 2020; Dodson et al., 2018b; Minnes et al., 2019; Stephenson, 2019; Weinert et al., 2020), analysis of video content (Wijnker et al., 2019; Lei et al., 2016; Ge & Li, 2020; Voronkin, 2019), own experiences (Hodges, 2018; Stephenson, 2019), interviews to teachers, assistants, and producers (Wijnker et al., 2019; Lei et al., 2016), analysis of the learning outcome (Minnes et al., 2019; Hung et al., 2018), and analysis of log data (Dodson et al., 2018b; Stephenson, 2019) and forum posts (Lei et al., 2016; Hung et al., 2018).

Reported impact on video effectiveness Controlled experiments often explored interactive features, followed by production styles, instructor behavior and cues, video length, completion, and pace, the difference between experts and beginners, and video popularity. Table 5 provides an overview of the effectiveness levels of the reviewed features, along with relevant details. These levels were determined based on reported experimental results, the number of studies supporting the findings, and our assessment of the potential of the feature. For instance, a higher effectiveness level was assigned when effects were positive and statistically significant, when multiple studies corroborated the findings, or when a feature demonstrated strong potential. However, in some cases, categorization does not directly apply. For example, certain features relate to learner performance rather than video content itself. While these features are not directly tied to the video, we considered them important to report. It is important to

Table 4 Overview of tools, tasks, and technologies developed to support VBL, along with their implementation frequency

Tool category	Tool subcategory	Implementation frequency	Studies that implemented the tool
Information extraction	Text	More frequent	Davila et al. (2021), Kota et al. (2020), Kota et al. (2019), Xu et al. (2019a), Lee et al. (2017b), Garber et al. (2017), Zhong and Ma (2018), Khattabi et al. (2019), Davila et al. (2021), Khattabi et al. (2019), Lee et al. (2017b), Davila and Zanibbi (2018), Xu et al. (2019a), Garber et al. (2017), Zhong and Ma (2018), Khattabi et al. (2019)
	Metadata	Less frequent	Jung et al. (2019), Escobar-Avila et al. (2017), Jr. et al. (2019), Furini (2018), Ghosh et al. (2021), Nangi et al. (2019)
	Key segments and topics	Less frequent	Ghauri et al. (2020), Che et al. (2016), Kravvaris and Kermanidis (2019), Lin et al. (2019), Liu et al. (2018), Ponzanelli et al. (2019)
Interactive tools	In-video annotations	More frequent	Fong et al. (2018), Liu et al. (2019), Malchow et al. (2018), Lai et al. (2018), Singh et al. (2016), Nguyen and Liu (2016), Deb et al. (2017)
	Quizzes	More frequent	Herault et al. (2018), Kohen-Vacs et al. (2016), Walk et al. (2020), Wachtler et al. (2016a), Ma and Ma (2019), Wachtler and Ebner (2019)
	Feedback	Less frequent	Lin and Chen (2019), Jo and Lim (2017), Robal et al. (2018), Sharma et al. (2016a), Llanda (2019)
	360-degree characteristics	Less frequent	Herault et al. (2018), Koumaditis and Chinello (2019), Berns et al. (2018)
Recommendation systems	Similarity-based (from transcripts and metadata)	More frequent	Zhao et al. (2018a), Waykar and Bharathi (2017), Bleoanca et al. (2020), Basu et al. (2016b), Jung et al. (2019), Ji et al. (2018), Jordán et al. (2020), Walk et al. (2020), Tavakoli et al. (2020)
	Concept-based (from ontologies and knowledge bases)	Less frequent	Schulten et al. (2020), Borges and dos Reis (2019), Coccoli and Vercelli (2016), Borges and Silveira (2019)
	Other: based on readability, demanded skills from labor market, popularity, instructor, learning style, pedagogical characteristics	Less frequent	Jung et al. (2019), Tavakoli et al. (2020), Kravvaris and Kermanidis (2017), Zhang et al. (2018a), Guan et al. (2019), Imran et al. (2019), Ciurez et al. (2019), Acuña-Soto et al. (2020)
Classifiers and predictors	Learning outcome prediction	More frequent	Lemay and Doleck (2020), Mbouzao et al. (2020), Mubarak et al. (2021), Furukawa et al. (2020), Korösi et al. (2018), Lallé and Conati (2020), Brinton et al. (2016)

Table 4 continued

Tool category	Tool subcategory	Implementation frequency	Studies that implemented the tool
	Discipline classification	Less frequent	Stoica et al. (2019), Dessì et al. (2017), Basu et al. (2016a), Othman et al. (2017), Chatbri et al. (2017)
	Instructor behavior prediction	Less frequent	Chen et al. (2020), Godavarthi et al. (2020), Sharma et al. (2016b), Xu et al. (2020)
	Dropout prediction	Less frequent	Jeon and Park (2020), Furini et al. (2021)
Navigation tools	Table-of-contents navigation	More frequent	Das and Das (2020), Kravvaris and Kermanidis (2019), Mahapatra et al. (2018a), Mukherjee et al. (2019), Husain and Meena (2019), Gandhi et al. (2016), Kumar et al. (2017), Zhao et al. (2017), Yadav et al. (2016), Bian and Huang (2019)
	Navigation by keywords	Less frequent	Ohnishi et al. (2019), Koka et al. (2020), Kravvaris and Kermanidis (2019), Kanadje et al. (2016), Kumar et al. (2017), Zhao et al. (2017)
	Navigation by visual elements	Less frequent	Zhao et al. (2018b), Ma et al. (2017), Kumar et al. (2017), Zhao et al. (2017), Yadav et al. (2016)
Analytic tools	Video usage (via play, pause, seek, etc.)	More frequent	Walk et al. (2020), Wachtler et al. (2016a), Long et al. (2018)
	Learners performance	More frequent	Min et al. (2019), He et al. (2018a), He et al. (2018b)
	Other: engagement and sentiment (via comment analysis)	Less frequent	Sung et al. (2016), Stepanek and Dorn (2017)
Summarization	From text (keywords, entities, subtitles, and metadata)	More frequent	Vimalaksha et al. (2019), Balasubramanian et al. (2016), Kim and Kim (2016)
methods	From visual content	Less frequent	Rahman et al. (2020)

note that the assigned levels serve only as suggestions, as the effectiveness of features is highly context-dependent.

Further trends and future work Our analyses corroborated **prior reviews'** findings: (1) we observed (2) an increase in the number of VBL research articles (Giannakos, 2013; Poquet et al., 2018), (3) a tendency on using learning outcome as a metric of learning effectiveness (Poquet et al., 2018; Yousef et al., 2014), (4) the next most popular metrics were cognitive load, satisfaction, mental effort, social presence, moti-

Table 5 Overview of impact of video characteristics on video effectiveness

Category feature	Subcategory feature	Effectiveness	Details
Interactive features	Quizzes	High	Effective in combination with other features and optimal timing (Kleftodimos & Evangelidis, 2018; Leisner et al., 2020; Wachtler et al., 2016b; Palaigeorgiou & Papadopoulou, 2019; Ding, 2018; da Graça Campos Pimentel et al., 2019; Kovacs, 2016)
	Virtual reality and 360-degree	Medium	Positive effects but without statistical significance (Chen, 2020; Araiza-Alba et al., 2021; Daher & Sleem, 2021; Huang et al., 2020)
	In-video annotations	High	Improved learning outcome (Leisner et al., 2020; Taskin et al., 2019; Kuhail & Aqel, 2020)
	Navigational elements	Medium	Improved learning outcome, but results were not always significant or mixed with effects of other features (Kuhail & Aqel, 2020; da Graça Campos Pimentel et al., 2019; Cojean & Jamet, 2017)
Production styles	Instructor's presence	Medium	Positive effects but not always significant (Pérez-Navarro et al., 2021; Davila et al., 2021; Hew & Lo, 2020)
	Animations	High	Higher learning outcomes compared to text media or traditional classes (Srivastava et al., 2020; Cookson et al., 2020; Al-Khateeb & Alduwairi, 2020)
	Tutorials vs. lectures	Medium	Both correlated positively with learning outcome (Hildebrand & Ahn, 2018)
	Dialogue vs. monologue	Low	No significant positive effects on learning outcomes (Ding et al., 2018; Nugraha et al., 2020; Lee & Muldner, 2020)
	Diverse styles	Medium	The use of diverse styles was found effective (Costley & Lange, 2017)
Instructor behavior	Speech characteristics	High	Impacted certain aspects such as understanding and attention (Atapattu & Falkner, 2017; Shoufan, 2019b; van der Sluis et al., 2016)
	Gestures	Medium	Improved aspects such as satisfaction and presence, but little effect on learning outcome (Beege et al., 2020; Wang et al., 2019a; Horovitz & Mayer, 2021)
	Instructor cues	Medium	Positive effect on learning outcome (Wang et al., 2020b; Pi et al., 2019)
	Content cues	Medium	Positively affected learning outcome, but can cause cognitive overload (Wang et al., 2020b; Pi et al., 2017b; Sharma et al., 2016c; Moon & Ryu, 2021)
	Animated instructors	Low	No positive effect (Sharma et al., 2016c; Moon & Ryu, 2021)

Table 5 continued

Category feature	Subcategory feature	Effectiveness	Details
Video length	Short videos	Medium	Preferred by learners in general (Saurabh & Gautam, 2019; Ding, 2018; Engeness & Nohr, 2019), but longer ones are preferred in some contexts (Giannakos et al., 2016; Shoufan, 2019a)
Video completion	Watching proportion	Medium	Improved learning outcome when watching beyond a certain proportion (Hildebrand & Ahn, 2018; Angrave et al., 2020)
	Skipping	Low	Common behavior that negatively affects video completion (Hildebrand & Ahn, 2018; Yan & Baxter, 2018; Giannakos et al., 2016)
Video pace	Automatic pause	Low	No significant effects. Some studies showed negative impacts on learning outcome (Schroeder et al., 2020; Garrett, 2018, 2021; Merkt et al., 2018)
	Faster speed	Medium	Beneficial in certain cases (Lang et al., 2020; Garrett, 2021)
Learners' performance	High achievers	-	Engaged in more diverse activity patterns compared to low achievers (Rodríguez et al., 2018; Kovacs, 2016)
	Specialists	-	Achieved higher learning outcomes and often engaged more in video activities than beginners (Li, 2019; Stöhr et al., 2019)
	Engaged	-	No difference in the learning outcome between more engaged and less engaged viewers (Galster et al., 2018)
Video popularity	Perceived helpfulness	Low	Learners liked videos they found helpful. However, high popularity did not always guarantee quality (Saurabh & Gautam, 2019; Lee et al., 2017a)
Other	Discipline, topic, content	-	Behavior changed according to the discipline, topics, and content of the video (Saurabh & Gautam, 2019; Uchidiuno et al., 2018; Hildebrand & Ahn, 2018)
	Platform	Low	YouTube videos favored over institutional ones (Giannakos et al., 2016)

vation, and self-efficacy (Poquet et al., 2018), (5) we confirmed that eye-tracking and log data characteristics were used; however, we also identified thermal (Srivastava et al., 2019, 2020), neurophysiological (Winkler et al., 2020), facial expressions-tracking (Srivastava et al., 2019) characteristics, and (6) a significant emphasis on STEM disciplines (Giannakos, 2013; Poquet et al., 2018), from where we can deduce that controlled experiments have yet to encompass the full breadth of subjects. Investigating the particularities of other topic areas, such as the humanities and arts, could be a fruitful endeavor.

Among our other results with regard to **controlled experiments**, we recognized slide-based videos as the most used **production style**, which implies that these studies have not tested the broad spectrum of video styles. However, there are indications that further insight could be gained by exploring more diverse productions beyond the easily available slide-based type. Research on style and layout aspects like handwriting, screencasts, and animated learning videos was, for example, rather limited. Note that Poquet et al. (2018) found that animated videos were popularly used; nonetheless, this could be due to the use of the keyword "animation" in their target search, which could have led to the retrieval of a higher number of articles using this particular presentation style. Furthermore, controlled experiments showed a strong focus on **short-term knowledge gain** as a metric of learning success. It seems as if long-term retention, skill acquisition, and transfer could be worthwhile concepts to study in more depth. Finally, controlled experiments often yielded **contradictory results** about the impact of video characteristics on learning effectiveness. The discrepancies can have multiple sources, among them, the diversity of test conditions, examined production styles, etc. We noticed that not all studies reported all characteristics of the investigated videos in sufficient detail to enable a direct comparison. Our comprehensive assessment of experimental research would have benefited from more descriptive reporting on the used videos and study participants. Therefore, we believe that further standardization and good practices in how results are informed are relevant aspects for improvement.

Technology-wise , we observe an uptake of the recently so popular **deep learning** techniques (as has other research in text and image analysis) for the development of tools for supporting learning. These techniques were used for basic tasks such as video segmentation, optical character recognition, neural word representations (embeddings), speech-to-text tasks, etc., and also in entire pipelines of recommendation systems, classifiers, predictors, etc. Although there has been progress, further enhancements are possible by integrating cutting-edge technologies into the realm of VBL. Additionally, we noticed that **video characteristics** were often analyzed by segregating a video's modalities with a predominant focus either on visual or auditory elements. However, video learning is achieved by the interaction of sound and images, as stated by the Cognitive Theory of Multimedia Learning (Mayer, 2021). Thus, we see a failure to handle videos as the *multimodal media* they are. First works in that direction exist (e.g. Shi et al., 2019), but presumably, novel multimodal models could bring a boost in performance for tools in the VBL area.

Implications for future AIED research The presented taxonomy of video characteristics provides an empirical grounding for the application of emerging AI technologies in educational contexts. While recent advances in large language models and generative AI have transformed the technological landscape, the evidence-based insights highlighted here can guide the integration of these powerful tools toward pedagogically sound implementations. The examined interactive features, learner activities, and instructional design principles offer specific parameters that AI systems can optimize when generating or adapting video content. Rather than developing AI-enhanced video tools based on technological capabilities alone, our findings enable researchers to target the specific video characteristics with proven learning impact. For example,

machine learning methods analyzing log data can now focus on the most predictive interaction patterns we have identified, and recommendation systems can prioritize videos with empirically validated characteristics matched to learner profiles. Additionally, future research could explore how these characteristics might help develop systems that strengthen learners' ability to evaluate source credibility in an era of growing misinformation. As AI capabilities expand, this taxonomy serves as a critical bridge between technological innovation and educational effectiveness-ensuring that advanced systems like adaptive content generators, intelligent video editors, and personalized learning environments are designed to implement the specific video features that demonstrably enhance learning outcomes.

Limitations We retrieved articles from academic computer science databases. Thus, we did not consider other disciplines that contributed to VBL research such as those from science education. We queried three dedicated databases, where one of them (DBLP) can be considered to track the vast majority of relevant CS venues (Dblp, 2025). We retrieved articles with the query terms only in the title to ensure that the main focus was on VBL – however, there might be articles treating VBL-related topics without it being reflected in the title. Moreover, we restricted our review to articles that used and experimented with video characteristics, thus, advances in other use cases such as teacher reflection were not reported. Finally, an accurate comparison of the effects of video characteristics on learning effectiveness was difficult due to the heterogeneity in the experimental settings and test conditions. This is why we could not provide a meta-analysis.

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Declarations

Competing interests The authors declare no competing interests.

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